

# FRC 2026 REBUILT — Top-30 Robot Design Deep Dive

A study reference for effective design & implementation, drawn from Chief Delphi, David's Design Server (frcdesign.org), team CAD/code releases, Behind the Bumpers, and Statbotics EPA.

## The 2026 design landscape

### The game

REBUILT is a fuel (6" foam ball) shooter: collect fuel and score it into your hub when active. Robots may hold unlimited fuel. The end-game tower climb existed but was LOW VALUE — most top teams dropped or never built a climber, so 2026 design is about the fuel cycle: intake → storage/indexing → launch.

### The defining meta

The season split into two offensive families: (1) high-volume 'DUMPERS' — huge hoppers that drive to the hub and empty fuel in bulk during scoring windows (1678 'Limestone', 254, 7769); and (2) 'SHOOTERS' — flywheel or drum launchers, frequently TURRETED with vision auto-aim for shoot-on-the-move (4414, 6329, 1987, 2767, 5000, 4481). Several strong teams REBUILT mid-season from a passer/multi-wide shooter to a dumper (2910 Blitz→ReBlitz; 9470 triple→drum; 5687 'Big Dumpy').

### Indexing was where teams innovated

Moving fuel by geometry beat squeezing it. Spindexers ('car wash' dual-stack on 1796, zero-compression drum on 6329, cone-in-bowl on 9483), drum/revolver magazines, and roller-floor serializers (1678, 1768) dominated. Large flow-through hoppers (50–80 fuel) were common.

### Drivetrain & vision

Swerve is universal at this tier. Vision was a real differentiator but was rarely published: confirmed stacks include 9470 (2× OV2311 global-shutter + Orange Pi 5 Pro, shoot-on-the-move derived from 6328), 1678 (Limelight 4 + CANrange), 3467 (custom Jetson + 4 cameras), 5000 (PhotonVision), 157 (PhotonVision on Orange Pi 5). Most teams left vision hardware unstated — marked 'unconfirmed' here, meaning not published, not absent.

### How to use this document

Profiles are ordered by Statbotics EPA (season-cumulative scoring contribution). For each team we capture archetype, the four subsystems, vision, public CAD/design availability, and 'lessons for students'. The teams with the richest reproducible documentation — best for a hands-on study — are 1678, 4414, 1690, 7769, 6329, 6800, 3467, 9470, 5000, and 4481.

## Top-30 at a glance (by Statbotics EPA)

| # | Team                  | Robot                 | EPA | Rec        | Archetype                                                      | Public CAD/Design |
|---|-----------------------|-----------------------|-----|------------|----------------------------------------------------------------|-------------------|
| 1 | * 4414 HighTide       | RIPCURRENT            | 357 | 70-2       | Single-turret flywheel shooter + 'dye-rotor' centering; high-B | Tech binder       |
| 2 | 254 The Cheesy Poofs  | Overload              | 328 | 65-6       | 'Overload': 4-wide DUMPER with an articulating hood. Chose dum | Team site         |
| 3 | * 7769 The CREW       | CHUNK                 | 312 | 70-5       | Wide single-stream 'dumper' (chose dumper over turret) with a  | CD thread         |
| 4 | 1323 MadTown Robotics | (undisclosed)         | 310 | 69-4       | Unconfirmed — strong results, minimal public documentation     | Stats/TBA         |
| 5 | 2056 OP Robotics      | (2026 binder pending) | 302 | 63-10      | Single-ball turret shooter fed by a high-throughput conveyor + | Video             |
| 6 | * 1690 Orbit          | KEPLER                | 295 | (district) | 'Kepler' trench-bot built on 4 pillars: shoot-on-the-move, lar | CAD release       |
| 7 | 27 Team RUSH          | (undisclosed)         | 295 | 83-6       | Wide 3-fuel STEEL-DRUM shooter fed by a roller floor + passive | Stats/TBA         |
| 8 | 1114 Simbotics        | Simbot Tim            | 288 | 62-7       | Wide DRUM dumper (deliberately NO turret); huge fuel capacity. | Team site         |

| #  | Team                           | Robot                  | EPA        | Rec   | Archetype                                                      | Public CAD/Design |
|----|--------------------------------|------------------------|------------|-------|----------------------------------------------------------------|-------------------|
| 9  | <b>2481 Roboteers</b>          | (undisclosed)          | <b>285</b> | 44-8  | 'Big dumper': fixed 4-wide shooter + 5-ball-wide touch-it-own- | Stats/TBA         |
| 10 | <b>5687 The Outliers</b>       | Glenn / 'Big Dumpy'    | <b>283</b> | 62-12 | Fast drum-shooter that iterated into a dumper ('Big Dumpy')    | Code (GitHub)     |
| 11 | <b>* 2910 Jack in the Bot</b>  | Blitz → ReBlitz        | <b>282</b> | 53-23 | Built a passer/3-wide shooter ('Blitz'), then REBUILT mid-seas | CAD release       |
| 12 | <b>125 NUTRONS</b>             | FLUX                   | <b>282</b> | 68-17 | 'Dye rotor' ('dish') shooter - NUTRONS originated the dye-roto | Team site         |
| 13 | <b>* 9470 Ctrl-Alt-Defeat</b>  | (triple → drum)        | <b>279</b> | 60-13 | Triple shooter rebuilt mid-season into a drum shooter; 'CTRL+A | Tech binder       |
| 14 | <b>7558 ALT-F4</b>             | (undisclosed)          | <b>279</b> | 51-17 | Drum shooter with an advanced vision/simulation software progr | Team site         |
| 15 | <b>* 1678 Citrus Circuits</b>  | Limestone              | <b>278</b> | 61-9  | World-record DUMPER / hooded shooter; led the 'Certified Dumpe | CAD release       |
| 16 | <b>6369 Mercenary Robotics</b> | (full-ramp intake bot) | <b>274</b> | 77-9  | Rebuild from a TURRET to a wide aluminum DRUM shooter; slapdow | Code (GitHub)     |
| 17 | <b>* 6800 Valor</b>            | Downpour               | <b>268</b> | 62-10 | Elevator-launched fuel scorer with assisted-aim and bidirectio | Onshape CAD       |
| 18 | <b>* 3467 Windham Windup</b>   | (Open Alliance)        | <b>266</b> | 75-10 | High-throughput dumper with auto-aim shooter backup; software/ | Onshape CAD       |
| 19 | <b>9128 ITKAN Robotics</b>     | Triple Threat          | <b>265</b> | 60-7  | Trench robot: cycloidal slapdown intake -> duct-tape conveyor  | Video             |
| 20 | <b>1796 RoboTigers</b>         | Sprout                 | <b>263</b> | 48-7  | 'Car Wash' spindexer fuel scorer                               | CD thread         |
| 21 | <b>9483 Overcharge</b>         | Enigma                 | <b>261</b> | 38-6  | Minimalist, low-motor-count spindexer scorer                   | Video             |
| 22 | <b>* 6329 The Bucks' Wrath</b> | Roman I / Roman II     | <b>260</b> | 61-15 | REBUILD story: 'Roman I' (turret + low-compression spindexer,  | Onshape CAD       |
| 23 | <b>2046 Bear Metal</b>         | Bearricade MkII        | <b>260</b> | 78-16 | 'Big dumper' with a hopper-floor + drum shooter; simplicity-ma | GrabCAD           |
| 24 | <b>1987 Broncobots</b>         | Cyclone                | <b>259</b> | 54-6  | 'Cyclone': turret-mounted DYE-ROTOR shooter (up to 720deg trac | Video             |
| 25 | <b>1768 Nashoba Robotics</b>   | Nightshift             | <b>259</b> | 70-17 | Fixed full-width high-throughput floor-loader; volume shooter, | CD thread         |
| 26 | <b>8044 Denham Venom</b>       | Cat-5                  | <b>257</b> | 50-9  | Ran two robots: v1 (low 'bump/under-trench) and v2 (a HighTid  | DDS thread        |
| 27 | <b>1706 Ratchet Rockers</b>    | (undisclosed)          | <b>256</b> | 45-2  | Dual-turret shooter ('Mirage' - 'shooting so fast you think yo | Code (GitHub)     |
| 28 | <b>9496 LYNK</b>               | Matterhorn             | <b>254</b> | 70-13 | 2-wide drum 'over-bumper reverse' shooter (modeled on 254's 20 | CD thread         |
| 29 | <b>5460 Strike Zone</b>        | Helios                 | <b>252</b> | 57-11 | Drum shooter mounted on a turret (a 'turret dumper'), ~30deg t | CD thread         |
| 30 | <b>* 581 Blazing Bulldogs</b>  | Rubble                 | <b>251</b> | 47-19 | 'Rubble': dye-rotor turret shooter w/ 35-deg adjustable hood,  | Onshape CAD       |

\* = recommended study pick (high performance + a usable public CAD or design release).

# Team-by-team profiles

|                 |                                                          |                 |
|-----------------|----------------------------------------------------------|-----------------|
| #1 (study pick) | 4414 HighTide <small>RIPCURRENT</small><br>San Diego, CA | EPA 357<br>70-2 |
|-----------------|----------------------------------------------------------|-----------------|

## Single-turret flywheel shooter + 'dye-rotor' centering; high-BPS single stream

|                                                                                                                                               |                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| SHOOTER<br>4× Kraken X44 on a 3" flywheel; turret geared 55.7:1 with a flex-wheel cable-chain tensioner; low gearing keeps launch current low | INTAKE<br>Over-bumper/ground intake, 2× Kraken X60 + 1× X44 bottom bar; left/right sides decoupled for durability                 |
| INDEXER / HOPPER<br>'Dye rotor' optimized for high-BPS single-stream feeding; structural bumpers double as hopper walls AND the shooter mount | DRIVETRAIN<br>25×32" swerve, 2 custom modules from WCP Swerve X2 parts, 7.67:1; rear corners chamfered to grow hopper volume      |
| ENDGAME<br>Climber not detailed in the binder.                                                                                                | AUTONOMOUS<br>High-BPS single-stream turret with vision auto-aim / shoot-on-the-move (controls-led; precomputed trajectory maps). |
| VISION<br>Limelight always pointed at the hub (reads the hub's 4 AprilTags), accounting for ~95% of on-robot localization [BtB transcript]    | PUBLIC CAD / DESIGN<br>Interactive tech binder (no downloadable CAD)                                                              |

### LESSONS FOR STUDENTS

- Make structure multi-functional: bumpers serve as hopper walls and shooter mount, saving weight and volume.
- Low flywheel gearing minimizes current spikes at launch — protects the whole electrical budget.
- Decoupling the two intake sides improves survivability against contact/defense.
- A polished, AI-assisted state-machine codebase + interactive binder is a model of documentation.

Tech binder: <https://2026.team4414.com/> · Tech binder · Reveal · Statbotics · Behind the Bumpers (full walkthrough)

**'Overload': 4-wide DUMPER with an articulating hood. Chose dumper over turret at their 12-hr kickoff build (turret ceiling ~18 BPS vs ~30-35 for a 4-wide multi-stream); robustness-first. ~25-28 BPS on fresh balls. Curie Division winner [BtB transcript]**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>SHOOTER</b><br/>4-wide dumper shooter with an articulating hood and a 'triangle of compression' - each ball contacts the main drum, the bottom feeder roller, and the top hood roller at once -&gt; zero bobble, full spin-up, consistent shots. Only ~1.25in clearance per side, so a unique DOUBLE-SIDED belt drives the main drum (outer) and the back hood rollers (inner) for the reversing effect; backing feeder swapped poly -&gt; CatTongue (poly slipped); shooter motors nested behind [BtB transcript]</p>                                                       | <p><b>INTAKE</b><br/>Most-iterated subsystem. Rack-and-pinion deploy: racks went from 3/8in polycarb w/ titanium teeth (kept breaking) to TWO layers of 1/4in SRP + one 1/8in titanium (unbreakable). A bottom 3-roller rips balls up TITANIUM kicker plates/bars (aluminum bent) tuned for low compression without dead spots; a 2nd roller keeps stored hopper balls from blocking incoming ones; dual Kraken X60. (Used GoPros inside to find jams; SRPP chosen to spread load around bolt holes, ~30% lighter, impact-resistant) [BtB transcript]</p> |
| <p><b>INDEXER / HOPPER</b><br/>'Bumper hopper' = a full 8in polycarb wall extending up from the bumpers (more width, fewer parts) + a carbon-fiber hoop/corners for rigidity; a NET raises capacity from ~50 to ~75-80 balls. Hopper floor = WCP rubber-back timing belts (smoother than rollers). The floor's pivot is LINKED to the intake deploy ('trash compacting'): pulling the intake in while shooting lifts the floor so gravity empties the last balls; the intake fully stows under the floor [BtB transcript]</p>                                                      | <p><b>DRIVETRAIN</b><br/>Swerve (module not detailed); deliberately does NOT shoot on the move (current limits) - instead optimizes for very fast alignment + a smart pivot point [BtB transcript]</p>                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>ENDGAME</b><br/>Climber not detailed (dumper archetype).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p><b>AUTONOMOUS</b><br/>Dumper: drive to the hub and empty fuel in bulk during scoring windows; auto routines not detailed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p><b>VISION</b><br/>FOUR cameras (their most ever) on ONE Jetson Orin Nano with GPU-accelerated AprilTag detection: two grayscale 120fps on the shooter side (heavy lifting) + two 60fps side cameras; MULTI-TAG only (rejects single-tag), ~100+ fps. The Jetson solves and sends poses over UDP to the RIO, which filters by reprojection error + ambiguity. Heading control uses a predicted look-ahead pose + feed-forwards + an intelligent center-of-rotation (pivots about a corner for fastest alignment); Choreo 3-stage autos with a recovery path [BtB transcript]</p> | <p><b>PUBLIC CAD / DESIGN</b><br/>No public CAD (reveal video + build blog only)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

#### LESSONS FOR STUDENTS

- 254's build blog shows disciplined prototyping (backspin single-wheel vs dual flywheel) before committing.
- Even the #2 team in the world withheld CAD — when a team publishes nothing, match video is your scouting source.
- Their strategic game-analysis method is as worth studying as the hardware.

**Team site:** <https://www.team254.com/first/2026/> · 254 2026 · [Reveal](#) · [DDS robot-pics-2026](#) · [Behind the Bumpers \(full transcript\)](#)

**Wide single-stream 'dumper' (chose dumper over turret) with a very high-capacity hopper**

|                                                                                                                   |                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Evolved from a two-chamber shooter to a 2.75"-wide shooter; praised for distance + accuracy     | <b>INTAKE</b><br>Linear slapdown intake; deploy arms cut from polycarbonate rack + 98A TPU printed chain tensioner; dynamic current limit (45A deploy → 5A hold) for compliance |
| <b>INDEXER / HOPPER</b><br>Very high capacity — ~70 fuel max, 50–60 in play; in/out throughput was their key edge | <b>DRIVETRAIN</b><br>Swerve (module unconfirmed)                                                                                                                                |
| <b>ENDGAME</b><br>Climber not detailed (dumper).                                                                  | <b>AUTONOMOUS</b><br>Drive-and-empty dumper cycle; auto routines not detailed.                                                                                                  |
| <b>VISION</b><br>Unconfirmed                                                                                      | <b>PUBLIC CAD / DESIGN</b><br>Public CAD + tech slides (Onshape)                                                                                                                |

**LESSONS FOR STUDENTS**

- Massive hopper capacity + high throughput beat turret complexity for them — a clear dumper case study.
- Candid failure analysis: polycarb intake teeth sheared every 2–3 matches at champs; they ran ~16 spares. Pick intake materials for fatigue.
- Software current-limit tuning can buy you mechanical compliance without extra hardware.

**CD thread:** <https://www.chiefdelphi.com/t/frc-7769-cad-tech-slides-chunk/521008> · [CAD + slides](#) · [Behind the Bumpers](#)

**Unconfirmed — strong results, minimal public documentation**

|                                                               |                                                                  |
|---------------------------------------------------------------|------------------------------------------------------------------|
| <b>SHOOTER</b><br>Unconfirmed                                 | <b>INTAKE</b><br>Unconfirmed                                     |
| <b>INDEXER / HOPPER</b><br>Unconfirmed                        | <b>DRIVETRAIN</b><br>Swerve (assumed)                            |
| <b>ENDGAME</b><br>Unconfirmed (minimal public documentation). | <b>AUTONOMOUS</b><br>Unconfirmed (minimal public documentation). |
| <b>VISION</b><br>Unconfirmed                                  | <b>PUBLIC CAD / DESIGN</b><br>No public CAD/design release       |

**LESSONS FOR STUDENTS**

- Strong results (#4, 69-4) don't require public documentation — design knowledge isn't always shareable.
- When a team publishes nothing, use TBA match video as the primary scouting source.
- A candidate to contact directly rather than study from public CAD.

**Stats/TBA:** <https://www.thebluealliance.com/team/1323/2026> · [Team site](#) · [TBA 2026](#)

## Single-ball turret shooter fed by a high-throughput conveyor + ~62-ball net hopper; world-record scorer

|                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Turret-mounted single-ball shooter: 2x Kraken X60 flywheel run at ONE fixed RPM (never varied up/down), hood on one Kraken X44; 370deg rotation via a sprung, non-bidirectional cable chain [BtB transcript]                                                                                                                                | <b>INTAKE</b><br>Linear EXTENDING intake with Lexan flex side-plates (survives collisions, enters opponents' zone); front rollers on a Kraken X60 with plain-rubber coating; surgical-tubing kicker bar added mid-season to stop balls sticking against the bumper; 3D-printed wire-management chain [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>Neoprene-coated 'hot dog' rollers + a live-bottom funnel; a bare-Lexan 'slip' roller before the conveyor as an anti-jam; conveyor = top+bottom timing belts (thicker WCP bottom belt) compressing balls up into the turret; a reverse-spinning anti-jam roller; net hopper (~62 fuel) repurposed from prior years [BtB transcript] | <b>DRIVETRAIN</b><br>Trapezoidal swerve base (per Behind the Bumpers)                                                                                                                                                                                                                                                 |
| <b>ENDGAME</b><br>Climber not detailed in the public material.                                                                                                                                                                                                                                                                                                | <b>AUTONOMOUS</b><br>Single-ball turret + high-throughput conveyor; auto/vision detail in their Behind the Bumpers video.                                                                                                                                                                                             |
| <b>VISION</b><br>Not named in the Behind the Bumpers walkthrough (throughput is conveyor-driven with a single-speed flywheel); camera/coprocessor unconfirmed                                                                                                                                                                                                 | <b>PUBLIC CAD / DESIGN</b><br>2025 binder public; 2026 not yet released                                                                                                                                                                                                                                               |

### LESSONS FOR STUDENTS

- 2056's 'Behind the Bumpers' series is a model for student-facing design explanation — watch the 2026 entry.
- Their prior technical binders (e.g. 2025) are an excellent template for a student design study.
- Research trap to avoid: 'LIGHTNING' is their 2025 robot, not 2026 — verify season before citing.

**Video:** <https://www.youtube.com/@2056public> · Team site · [YouTube](#) · TBA 2026 · [Behind the Bumpers \(read full transcript\)](#)

**'Kepler' trench-bot built on 4 pillars: shoot-on-the-move, large (~60-ball) hopper, extreme durability, low profile; roller-floor indexer into a 640deg turret. Galileo Division winner [BtB transcript]**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Compact 640deg turret (360+ per side): X60 drive through a low gearbox (low CG), self-made gears on an 8in bearing, ball path offset toward the back to tuck into the corner. Shooter tuned for slight/no spin (most consistent stream, lowest energy loss): back rollers belted to a custom main shooter wheel; motors on an SLS winch pulled tight to the wheel + hood motors on the plate side via bevel gears to MINIMIZE diameter; custom 3D-printed energy chain + 14ga wires + a raised 'neck' so the chain wraps tight enough for 640deg [BtB transcript]                                                            | <b>INTAKE</b><br>Durable/collapsible/swappable wide intake: two silicone-coated rollers on a Kraken X60 + a static bottom kicker roller; deliberately NO slapdown (poor impact collapse) - instead an in-house RACK-AND-PINION deploy (polycarb racks + aluminum/brass gears); 25x50mm front extrusion + a polycarb/aluminum side sandwich; the whole U-frame lifts off via 3 screws per side for fast swaps [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>3 stages: (1) roller floor of 11 carbon-fiber silicone rollers on an X60 (2:1), roller-to-roller via 3mm GT2 belts, a slight reduction to ease the 4-wide -> 3.5-wide jam point, plus two slick (un-coated) rollers to cut friction; (2) two 30mm 5mm-HTD vertical belts on two independent X44 (~8 m/s) over a 1mm PTFE sheet (~10mm compression) + a vertical kicker roller (repurposed from their 2024 Doppler shooter) on an X44 to clear jams; (3) an active 'C' shooter ramp (3 rows of 2.25in flex wheels + 2 rows of 2in stealth wheels) on an X60 -> 8 m/s with ZERO spin into the turret [BtB transcript] | <b>DRIVETRAIN</b><br>29.5x24.5in (wide side = intake); bumpers attach to the hopper walls (thin polycarb + a net for ~60-ball capacity). MID-SEASON swap from a custom MK5-based module to stock SDS MK5 modules after the defense bot pushed them around - the MK5's wide spike treads dig into the carpet vs their thin nitro treads; reinforced/integrated top plates. 'Defense mode' X-locks the swerve [BtB transcript]  |
| <b>ENDGAME</b><br>Low-profile 'trench-runner' geometry trades climb access for faster, safer collection — climbing is not a focus.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>AUTONOMOUS</b><br>Shoot-on-the-move is one of the four stated design pillars (see CAD/reveal).                                                                                                                                                                                                                                                                                                                             |
| <b>VISION</b><br>A camera ON THE TURRET on an Orange Pi running PhotonVision (aimed at the hub's AprilTags) + two rear cameras for blind spots when the turret cable wraps or a camera drops; tilt + bump detection lets it shoot while tilted; an off-robot shoot-on-the-move simulation (gravity, air resistance, Magnus) precomputes trajectory polynomials the robot evaluates live [BtB transcript]                                                                                                                                                                                                                                       | <b>PUBLIC CAD / DESIGN</b><br>Full public CAD (Onshape + x_t)                                                                                                                                                                                                                                                                                                                                                                 |

**LESSONS FOR STUDENTS**

- Choosing a low-profile/trench geometry trades climb access for faster, safer ball collection.
- First full year on Onshape — a good case study in CAD-tool migration and clean release practice.
- A full-floor roller intake maximizes pickup area and tolerates sloppy approaches.

**CAD release:** <https://www.chiefdelphi.com/t/frc-orbit-1690-2026-robot-cad-release/520673> · [CAD release](#) · [Reveal 'Kepler'](#) · [Video](#) · [Behind the Bumpers \(full transcript\)](#)

## Wide 3-fuel STEEL-DRUM shooter fed by a roller floor + passive hopper; high-throughput dumper. Hall of Fame; Michigan State champion; ~330 EPA, 113 lb [BtB transcript]

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>One main STEEL 3in drum run by 4x Kraken X60 + two 1in hood rollers on the same X60s (spin ~2x faster to minimize backspin); 1x X44 rack-and-pinion hood; mostly plastic SRP + carbon-fiber structure for weight. VARIABLE compression (aluminum rods ~5/8in, CF stability rods ~3/8in, final ~1/2in) to fit a retrofit shooter compactly; the feeder accelerates fuel BEFORE the shooter, so the shooter only adds the final boost [BtB transcript] | <b>INTAKE</b><br>Main roller on 2x Kraken X44 (2.5:1) + a free-spinning carbon-fiber top roller + 3 passive inner tubes; built for quick fixes (it's at the robot's fighting face); the HOPPER is driven PASSIVELY off the intake [BtB transcript]         |
| <b>INDEXER / HOPPER</b><br>Roller floor + indexer on 2x Kraken X60; floor = 1in polycarb rollers (5 with CatTongue tape) funneling to the 3-wide shooter via dead-axle 3D-printed guides; geared 1:1 in front, ~17:19 at the back to speed balls up; a free-spinning roller prevents jams (replaced a belted one). Iterated from belt/sushi/chain to all-polycarb rollers [BtB transcript]                                                                             | <b>DRIVETRAIN</b><br>Swerve; ALL electronics mounted on the bottom (frees space for the roller floor + shooter); the electrical cover attaches to the BUMPERS so its weight doesn't count against inspection; 3D-printed component mounts [BtB transcript] |
| <b>ENDGAME</b><br>Unconfirmed — reveal is announcement-only.                                                                                                                                                                                                                                                                                                                                                                                                           | <b>AUTONOMOUS</b><br>Unconfirmed — reveal is announcement-only.                                                                                                                                                                                            |
| <b>VISION</b><br>Main Limelight 4 (heavily weighted, accepts tags mainly from the rear given the shooting orientation) + left & right Limelights fused as one entity into the pose estimate for odometry during feeding / when the back can't see [BtB transcript]                                                                                                                                                                                                     | <b>PUBLIC CAD / DESIGN</b><br>No public CAD/design release                                                                                                                                                                                                 |

### LESSONS FOR STUDENTS

- An 83-6 record (highest win count in the top-30) with no public release: execution and consistency can outweigh documentation.
- Reveal video + match footage are the only primary design sources.
- Michigan's competitive density (27, 7769, 5460...) is itself a design driver — strong regions force strong robots.

Stats/TBA: <https://www.thebluealliance.com/team/27/2026> · [Reveal thread](#) · [Video](#) · TBA · [Behind the Bumpers \(full transcript\)](#)

## Wide DRUM dumper (deliberately NO turret); huge fuel capacity. Hall of Fame; won both Ontario districts + the Ontario provincial championship [BtB transcript]

|                                                                                                                                                                                                                                       |                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Big belt-driven DRUM, an X44 + 3 more motors; hood uses only ~8deg of travel. They chose a drum (more versatile) OVER a turret (didn't suit their play style) and chose not to go under the trench [BtB transcript] | <b>INTAKE</b><br>2x Kraken X60; balls hit a kicker bar + main roller; an 'agitating' intake with multiple states - it can vibrate fast to push balls into the indexer harder, or run gentler [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>Extending hopper -> a moving 'rolling floor' (two rollers) carries balls up to the drum [BtB transcript]                                                                                                   | <b>DRIVETRAIN</b><br>Swerve (not detailed in the walkthrough)                                                                                                                                                 |
| <b>ENDGAME</b><br>Climber not detailed (wide drum dumper, no turret).                                                                                                                                                                 | <b>AUTONOMOUS</b><br>Drive-and-empty dumper; auto routines not detailed.                                                                                                                                      |
| <b>VISION</b><br>3x Limelight cameras with MegaTag2 + AprilTags, fused with odometry for pose estimation [BtB transcript]                                                                                                             | <b>PUBLIC CAD / DESIGN</b><br>No 2026 CAD release confirmed                                                                                                                                                   |

### LESSONS FOR STUDENTS

- Simbotics is well-known for STRATEGIC DESIGN EDUCATION — their resources/presentations are a top study reference even without 2026 CAD.
- Watch the 'Simbot Tim' Behind the Bumpers for subsystem rationale.
- Pair with 2056 (also Ontario) to compare two strong programs' 2026 approaches.

Team site: <https://www.simbotics.org/resources> · [Showcase](#) · [Behind the Bumpers](#) · [Resources](#)



## 'Big dumper': fixed 4-wide shooter + 5-ball-wide touch-it-own-it intake + 115-ball hopper (no turret); Milstein Division winners [BtB transcript]

|                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                           |
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| <b>SHOOTER</b><br>4-wide fixed shooter, ~20 BPS; 4 accelerator rollers for accuracy; 3x Kraken X44 on the shooter + 2x X44 + 1x X60 on feed rollers; grip-tape (skateboard tape) for consistent ball spin (+~40% accuracy), a late-season PVC-pipe guide (+~10%); 2-position linear-actuator hood (replaced a pneumatic piston to cut weight), flywheel RPM varied for distance [BtB transcript] | <b>INTAKE</b><br>5-ball-wide 'touch-and-take' intake: 3in roller wrapped in grip tape on 2x Kraken X60, SRPP side plates with 'Flintstone' wheels; ball compressed between roller and ramp; anti-jam = a 2nd motor + a plate to stop the roller stalling [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>Massive 115-ball, 5-wide hopper funneling to the 4-wide shooter through the accelerator rollers [BtB transcript]                                                                                                                                                                                                                                                      | <b>DRIVETRAIN</b><br>Custom swerve (designed in the 2024 offseason) with very large wheels and an adjustable ratio (12-32 ft/s); geared LOW this year for torque/low current under the heavy fuel load [BtB transcript]                                                   |
| <b>ENDGAME</b><br>Climber not detailed (fixed 4-wide dumper, no turret).                                                                                                                                                                                                                                                                                                                         | <b>AUTONOMOUS</b><br>Dumper; auto routines not detailed.                                                                                                                                                                                                                  |
| <b>VISION</b><br>Not mentioned in the Behind the Bumpers walkthrough; camera/coprocessor unconfirmed                                                                                                                                                                                                                                                                                             | <b>PUBLIC CAD / DESIGN</b><br>No public 2026 release (GitHub stops at 2025)                                                                                                                                                                                               |

### LESSONS FOR STUDENTS

- Strong record (44-8) with a low public footprint — another reminder strong results needn't be open-sourced.
- Study via match video; consider contacting the team directly.

**Stats/TBA:** <https://www.thebluealliance.com/team/2481/2026> · Team site · TBA 2026 · [Behind the Bumpers \(full transcript\)](#)

## Fast drum-shooter that iterated into a dumper ('Big Dumpy')

|                                                                                                                                                                 |                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>'Insanely quick' near-full drum shooter; high cycle rate prioritized                                                                          | <b>INTAKE</b><br>Over-bumper roller intake — 2" upper + 1.25" lower roller with ~½" compression (minimized to avoid bogging the motor) |
| <b>INDEXER / HOPPER</b><br>CTRE/Kraken 'Feeder' subsystem feeding a calculated-shot ('ShotCalculator') shooter; 3-roller intake (top/bottom rollers + deployer) | <b>DRIVETRAIN</b><br>Swerve (team has historically experimented with differential swerve)                                              |
| <b>ENDGAME</b><br>Climber not detailed (drum -> dumper).                                                                                                        | <b>AUTONOMOUS</b><br>Fast drum/dumper cycle; auto routines not detailed.                                                               |
| <b>VISION</b><br>Hybrid Limelight + PhotonVision: 3 cameras (front + left + right), AprilTag pose (MegaTag); C++ codebase                                       | <b>PUBLIC CAD / DESIGN</b><br>Code public (GitHub, C++); no CAD release                                                                |

### LESSONS FOR STUDENTS

- Empirical compression tuning — they minimized roller compression by experiment to reduce motor load.
- A separate 'LifeBot/Big Dumpy' repo signals a deliberate mid-season pivot to the dumper meta.
- A C++ codebase (less common than Java) is a good study for advanced programmers.

**Code (GitHub):** <https://github.com/frc5687> · 'Glenn' reveal · GitHub · TBA · [Robot code \(C++\)](#)

## Built a passer/3-wide shooter ('Blitz'), then REBUILT mid-season into a lightning dumper ('ReBlitz')

|                                                                                                                                                   |                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Blitz: 3-fuel-wide shooter, whole robot aims like a 3-wide turret, ~30 fuel/sec; ReBlitz dumper hit 30+ balls/sec               | <b>INTAKE</b><br>Dual intakes on Blitz                                                            |
| <b>INDEXER / HOPPER</b><br>Blitz used a multilayered indexer + intentionally SMALL hopper (a noted tradeoff); ReBlitz went to an open dumper path | <b>DRIVETRAIN</b><br>Swerve (long in-house swerve lineage)                                        |
| <b>ENDGAME</b><br>Climber not detailed (mid-season rebuild to a dumper).                                                                          | <b>AUTONOMOUS</b><br>Dumper after the Blitz -> ReBlitz rebuild; auto routines not detailed.       |
| <b>VISION</b><br>Unconfirmed for 2026                                                                                                             | <b>PUBLIC CAD / DESIGN</b><br>Onshape CAD + tech binders for BOTH robots (CD release, 2026-06-07) |

### LESSONS FOR STUDENTS

- THE case study in the season's central lesson: they scrapped a working passer and rebuilt as a dumper — and it won PNW DCMP + a return to Einstein.
- Now fully studyable: downloadable Onshape CAD AND a technical binder were released for BOTH Blitz and Re-Blitz, so you can diff the original vs the rebuild directly.
- Explicitly shows the hopper-capacity vs throughput tradeoff.
- Demonstrates the courage/value of abandoning a working robot mid-season when the meta shifts.

CAD release: <https://www.chiefdelphi.com/t/2910-cad-and-tech-binder-release-2026/521705> · CAD + Tech Binder release · Blitz reveal · Behind the Bumpers

## 'Dye rotor' ('dish') shooter - NUTRONS originated the dye-rotor concept (2017) and went BIGGER this year; trapezoidal drive base [BtB transcript]

|                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Enlarged dye-rotor 'dish' (sized up this year for throughput) fed by a live-shaft 3in carbon-fiber drum (CatTongue tape + silicone wrap) with a 3-standoff kick bar that maintains ball compression through the path [BtB transcript] | <b>INTAKE</b><br>Four-bar linkage deploy, modular (whole intake pops off via 2 shoulder bolts per side); a stiffener bar for aggressive driving; the intake is carved out so the robot can sit UNDER the trench at match start [BtB transcript]                                                           |
| <b>INDEXER / HOPPER</b><br>Live-shaft 3in CF drum + 3-standoff kick bar feeding the dye-rotor dish [BtB transcript]                                                                                                                                     | <b>DRIVETRAIN</b><br>Trapezoidal base - wider at the FRONT (bigger intake) and narrower at the REAR (balls rarely reach the back); 1/4in side plates (no 2x1 main structure); 2 stock WCP Swerve X2S modules at the front + 2 custom-machined X2S at the rear, shaped to the dish radius [BtB transcript] |
| <b>ENDGAME</b><br>Climber not detailed (dye-rotor shooter).                                                                                                                                                                                             | <b>AUTONOMOUS</b><br>Dye-rotor shooter; auto routines not detailed.                                                                                                                                                                                                                                       |
| <b>VISION</b><br>Not mentioned in the Behind the Bumpers walkthrough; camera/coprocessor unconfirmed                                                                                                                                                    | <b>PUBLIC CAD / DESIGN</b><br>No 2026 CAD release located                                                                                                                                                                                                                                                 |

### LESSONS FOR STUDENTS

- A 68-17 record from a distributed 5-school co-op is a model for sustaining strong performance across a large org.
- Mechanism detail lives in the FUN 'Behind the Bumpers' episode — watch rather than read.

Team site: <https://www.nutrons.com/> · Reveal thread · Behind the Bumpers · TBA

## Triple shooter rebuilt mid-season into a drum shooter; 'CTRL+ALT+DEL' deadline-driven scope control

|                                                                                                                                                                                                            |                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Drum shooter (mid-season rebuild from a Week-1 triple shooter): open fuel path, three feeder rollers on two Kraken X60.                                                                  | <b>INTAKE</b><br>Reinforced beater-bar intake — a flippable 2x1 aluminum bar (flip it when bent), then two 0.25" SRP leaf-spring plates per side with the intake tube as a hardstop and a detachable roller mount. |
| <b>INDEXER / HOPPER</b><br>Hopper with COTS aluminum hubs + compliant floor wheels carried over from the triple-shooter robot.                                                                             | <b>DRIVETRAIN</b><br>Swerve; a CANivore drives 18-motor shoot-on-the-move bursts.                                                                                                                                  |
| <b>ENDGAME</b><br>Climber not detailed — engineering hours went to scoring and intake robustness.                                                                                                          | <b>AUTONOMOUS</b><br>Empirical shot map (measured, not modeled; first-shot +250 RPM compensation from data) with shoot-on-the-move.                                                                                |
| <b>VISION</b><br>Two OV2311 global-shutter cameras + Orange Pi 5 Pro coprocessors; SOTM / vision / current-limiting adapted from 6328 and the NINJAS; 4.6 V brownout floor so failures degrade gracefully. | <b>PUBLIC CAD / DESIGN</b><br>Excellent public tech binder + GitHub                                                                                                                                                |

## LESSONS FOR STUDENTS

- Their 'CTRL+ALT+DEL' rule ships the robot 1.5 weeks before every event — the deadline decides what to cut, copy, or skip (applied 3x: build, rebuild, reinforce).
- Reinforce the proven failure point (the intake) instead of chasing optimization — robustness over speed for playoffs.
- Agentic coding tools for boilerplate let juniors plan instead of writing Java from scratch.

**Tech binder:** <https://9470techbinder.vercel.app/> · [Tech binder](#) · [Engineering process](#) · [Behind the Bumpers](#)

## Drum shooter with an advanced vision/simulation software program; rebuilt after their 2nd event; Ontario Technology Division champions [BtB transcript]

|                                                                                                                                                                                                                                                                                      |                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>3in aluminum DRUM with silicone tubing + two 1.5in silicone hood rollers; drum powered by 4x Kraken X60 geared 1:1; plus a custom laser-cut 1/2in-thick x 4in-wide STEEL flywheel [BtB transcript]                                                                 | <b>INTAKE</b><br>Linear intake running on HTD parts with 10DP gear-tooth racks meshing 10DP gears [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>Unconfirmed                                                                                                                                                                                                                                               | <b>DRIVETRAIN</b><br>Swerve (not detailed in the walkthrough)                                                      |
| <b>ENDGAME</b><br>Climber not detailed.                                                                                                                                                                                                                                              | <b>AUTONOMOUS</b><br>Advanced vision/simulation software program drives auto-aim; routines in code.                |
| <b>VISION</b><br>Multiple Limelights with a sophisticated localization pipeline: per-camera outlier rejection -> 'smart' MegaTag1/MegaTag2 selection via hysteresis -> staged standard-deviation calc; logs everything to simulate the robot without owning a field [BtB transcript] | <b>PUBLIC CAD / DESIGN</b><br>No public 2026 release located                                                       |

## LESSONS FOR STUDENTS

- Their 2024 turret-on-swerve + efficient diverter suggests strong mechanism-aiming experience.
- 2026 design is video-only — treat as a watch-the-reveal study.

**Team site:** <https://www.team7558.com/> · [Team site](#) · [YouTube](#) · [TBA](#) · [Behind the Bumpers \(full transcript\)](#)

## World-record DUMPER / hooded shooter; led the 'Certified Dumper Group' with 254

|                                                                                                                                                                   |                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Hooded flywheel fed from a full-width 3.5" aluminum drum (grip-taped); hood on an X44 jackshaft for a wide shot-angle range                     | <b>INTAKE</b><br>Full-width slapdown ground intake — a ~6 lb aggressive over-bumper roller (Kraken X60 pivot, silicone roller + kicker bar) |
| <b>INDEXER / HOPPER</b><br>Roller floor (dead-axle + flex-wheel rollers) → ball tunnel/serializer; expanding hopper; pragmatic 'strategically placed wood' guides | <b>DRIVETRAIN</b><br>Swerve, SDS MK5n, 27×27"                                                                                               |
| <b>ENDGAME</b><br>Climber not detailed (world-record dumper / hooded shooter).                                                                                    | <b>AUTONOMOUS</b><br>Drive-and-empty dumper; auto routines not detailed.                                                                    |
| <b>VISION</b><br>Limelight 4 (single rear unit) + 2× CANrange sensors; custom localization + a driver dashboard with a 'shift' system (CONFIRMED)                 | <b>PUBLIC CAD / DESIGN</b><br>Full CAD + code (Onshape, GitHub)                                                                             |

### LESSONS FOR STUDENTS

- The reveal that 'confirmed the dumper meta' — a community thread analyzed what Limestone implied about REBUILT strategy.
- Localization + a custom driver dashboard show software/driver UX as a force multiplier.
- Pragmatic 'wood in the ball path' — strong teams still use simple, cheap fixes.
- Fully open CAD + code makes this the single best reproducible study target of the season.

**CAD release:** <https://www.chiefdelphi.com/t/1678-citrus-circuits-2026-cad-and-robot-code-release/521535> · [Reveal: Limestone](#) · [Onshape CAD](#) · [Behind the Bumpers](#)

## Rebuild from a TURRET to a wide aluminum DRUM shooter; slapdown kicker-ramp intake → rolling floor → drum; net hopper (~78 fuel). Won Amarillo/Waco/San Antonio + Impact + EI [BtB transcript]

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                |
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| <b>SHOOTER</b><br>3in aluminum DRUM (only ~1/6in wide → too little momentum, so they added a 1.5 lb WCP STAINLESS flywheel at 1:3 - stainless beat brass, which wore down); drum on 4x Kraken X60 at 1:1; shooter geared 60:1 on a single X44 (down from 400:1); transfer = two 2in + one 1.25in rollers tilted back to passively funnel; back rollers two 1.25in silicone; ~20 ft/s for 60-70 balls; all motors packaged under the drum. Chose drum over turret for higher throughput + simpler/more-reliable software [BtB transcript] | <b>INTAKE</b><br>Slapdown intake (kept from robot 1) with a full KICKER RAMP (vs a kicker bar) for max fuel contact time + sustained compression (Citrus-style push into the indexer); 3 bars lift fuel high so the rolling floor can empty the hopper without much agitation [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>Rolling floor of custom 3D-printed stub rollers linked by belts (the intake jackshaft is hidden inside a roller to use the full width; one roller pivots with the intake); hopper extends exactly 12in + a net → ~50-55 fuel un-expanded, up to ~78 at 30in with the net up (for bump-overfill cycles) [BtB transcript]                                                                                                                                                                                       | <b>DRIVETRAIN</b><br>Swerve - swapped WCP X2 → SDS MK5N (easy to replace), geared for speed + torque [BtB transcript]                                                                                                                                                                          |
| <b>ENDGAME</b><br>Climber not detailed (turret → drum rebuild).                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>AUTONOMOUS</b><br>Auto routines not detailed; chose drum for simpler, more-reliable software.                                                                                                                                                                                               |
| <b>VISION</b><br>Three Limelight 4s (two front, one back) giving full-field localization in teleop + auto; one-button point-and-drive aiming with X-lock against defense; auto-aim auto-adjusts during autos [BtB transcript]                                                                                                                                                                                                                                                                                                            | <b>PUBLIC CAD / DESIGN</b><br>Code on GitHub; no confirmed 2026 CAD                                                                                                                                                                                                                            |

### LESSONS FOR STUDENTS

- Best win-rate in the cohort (77-9) despite a lower EPA ceiling — a case study in consistency and alliance leadership over raw scoring.
- 'Full ramp intake' points to a wide, forgiving ground pickup optimized for reliability.

**Code (GitHub):** <https://github.com/mercs6369> · [Behind the Bumpers](#) · [GitHub](#) · [DDS robot-pics-2026](#)

**Elevator-launched fuel scorer with assisted-aim and bidirectional shooting**

|                                                                                                                                             |                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Dual carbon-fiber arm launcher fed up a ~47" elevator (~0.3 s travel); can shoot in two directions                        | <b>INTAKE</b><br>Rotating intake synchronized with the arm for hand-off                                |
| <b>INDEXER / HOPPER</b><br>Internal feeder routing fuel intake→launcher; designed for fast maintenance access                               | <b>DRIVETRAIN</b><br>Swerve                                                                            |
| <b>ENDGAME</b><br>Climber not detailed.                                                                                                     | <b>AUTONOMOUS</b><br>Assisted-aim, bidirectional elevator-launched scorer; auto routines not detailed. |
| <b>VISION</b><br>Top-mounted camera for assisted scoring/alignment with a reverse-kinematic software model (published in their tech packet) | <b>PUBLIC CAD / DESIGN</b><br>Public CAD (Onshape)                                                     |

**LESSONS FOR STUDENTS**

- One living CAD assembly iterated all season (branches per version) keeps design history in one place.
- Maintenance-driven layout (top electronics, bottom battery plate, zip tie battery-lead hooks) — design for the pit, not just the field.
- An elevator-launch architecture is a creative alternative to flywheel/turret shooters.
- Published a full Tech + Engineering-Ops binder — strong documentation discipline.

**Onshape CAD:** <https://cad.onshape.com/documents/15cd41462dc24ff986a4e549> · [Tech binder \(PDF\)](#) · [CAD release thread](#) · [Reveal](#) · [Behind the Bumpers](#)

**High-throughput dumper with auto-aim shooter backup; software/autonomous-led**

|                                                                                                                       |                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Primarily a bulk dumper (fill hopper, empty during on-shifts) with an auto-aim shot vs tower/trench | <b>INTAKE</b><br>Linear-slide retracting intake with polycarbonate hopper walls; tested 1- vs 2-motor intake |
| <b>INDEXER / HOPPER</b><br>Large hopper + indexer/tower rollers; tuned indexer floor angle and speed for throughput   | <b>DRIVETRAIN</b><br>Swerve on Thrifty Narrow modules (switched from MK4i), CANivore/CANcoder, Choreo paths  |
| <b>ENDGAME</b><br>Climber not detailed.                                                                               | <b>AUTONOMOUS</b><br>Software/autonomous-led: a bulk dumper with an auto-aim shooter backup.                 |
| <b>VISION</b><br>Custom NVIDIA Jetson + 4-camera setup (CONFIRMED); AprilTag localization                             | <b>PUBLIC CAD / DESIGN</b><br>Full public CAD (Onshape) + Open Alliance blog                                 |

**LESSONS FOR STUDENTS**

- Autonomous CONSISTENCY beats peak score — they were passed at alliance selection for inconsistent autos, then fixed odometry (lowered Choreo friction coeff, restored drivetrain current limits).
- New-carpet and CAN-terminator gremlins caused real losses — document environmental variables.
- Their Open Alliance blog is a model of transparent post-event self-critique.
- Deliberate 'simple week-1 bot, planned mid-season upgrade' scheduling.

**Onshape CAD:** <https://cad.onshape.com/documents/7ef2c49c5bb48212a6b180ee> · [Open Alliance blog](#) · [CAD](#)

## Trench robot: cycloidal slapdown intake -> duct-tape conveyor -> transfer -> drum shooter, with a passive 'hopper flopper'. Won Space City 1 + Farmersville + an Impact award [BtB transcript]

|                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Drum shooter: 2x Kraken X60 on a ~13 lb flywheel mass (steel flywheels + wheels); BANG-BANG control (from their FTC shooter experience) to ramp fast, an operator pre-ramp button while crossing the bump, and a VARIABLE transfer speed (slows/speeds vs the flywheel velocity-vs-setpoint so each ball takes max energy); ~25-30 BPS [BtB transcript]                 | <b>INTAKE</b><br>Flip-down SLAPDOWN intake on a CYCLOIDAL pivot drive (Kraken X60 -> cycloidals; chosen for ease of use + better ball compression than a linear intake); two rollers (3in hard wheels + a roller bar) on 2x Kraken X44; a pin-and-slot links the intake raise to pull the hopper back in (agitation) [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>7-roller conveyor with DUCT-TAPE flaps to fling balls into the transfer (the tape jumped BPS from ~18-20 to ~25-30); transfer = three 1in rollers + bottom stationary tubes for compression, rollers on slots for tuning. Passive 'hopper flopper' - a flexible hopper that collapses to go under the trench and re-expands to store ~80 fuel [BtB transcript] | <b>DRIVETRAIN</b><br>Swerve (not detailed); built as a trench robot [BtB transcript]                                                                                                                                                                                                                                                  |
| <b>ENDGAME</b><br>Climber not detailed.                                                                                                                                                                                                                                                                                                                                                   | <b>AUTONOMOUS</b><br>Drum shooter; auto routines not detailed (Impact award).                                                                                                                                                                                                                                                         |
| <b>VISION</b><br>A single Limelight facing the shooting direction; planned a 2nd but used software instead - an auto 'point-towards' the opposing hub to update odometry before crossing the bump (so it doesn't hit the trench) [BtB transcript]                                                                                                                                         | <b>PUBLIC CAD / DESIGN</b><br>No public CAD (reveal + Behind the Bumpers)                                                                                                                                                                                                                                                             |

### LESSONS FOR STUDENTS

- A drum (revolver-style) shooter trades flywheel spin-up for very high BURST rate — strong for empty-the-hopper volleys.
- Distinguish BURST throughput (20–25 BPS) from SUSTAINED (15–16 BPS) when specifying a shooter; they are different targets.
- Running two near-identical robots (9128/10340) multiplies build & test learning.

**Video:** <https://www.youtube.com/watch?v=5HtLpv4AtIE> · Reveal 'Triple Threat' · Behind the Bumpers

## 'Car Wash' spindexer fuel scorer

|                                                                                                                                                                           |                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Launcher fed from a spindexer (launch type unconfirmed)                                                                                                 | <b>INTAKE</b><br>Intake pre-fills/'primes' the feeder while the outer spindexer stack turns slowly |
| <b>INDEXER / HOPPER</b><br>'Car Wash' spindexer — two independently driven stacks, each on its own Kraken; outer stack spins slow during intake to evenly pack the hopper | <b>DRIVETRAIN</b><br>Swerve (specifics unconfirmed)                                                |
| <b>ENDGAME</b><br>Climber not detailed.                                                                                                                                   | <b>AUTONOMOUS</b><br>Auto routines not detailed ('Car Wash' spindexer).                            |
| <b>VISION</b><br>Unconfirmed                                                                                                                                              | <b>PUBLIC CAD / DESIGN</b><br>No public CAD (photos on request)                                    |

### LESSONS FOR STUDENTS

- Independently driven indexer stages (separate motors) let you intake and prime simultaneously without jamming.
- Slow-spin-while-intaking evenly distributes fuel — a simple control trick for hopper packing.
- Cited by peers as 'one of the best robots in the world' despite no public CAD.

**CD thread:** <https://www.chiefdelphi.com/t/robotigers-team-1796-presents-sprout-2026/516769> · Reveal 'Sprout'

## Minimalist, low-motor-count spindexer scorer

|                                                                                                                              |                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| SHOOTER<br>Launcher fed from a spindexer (type unconfirmed)                                                                  | INTAKE<br>Two top rollers + a bumper-level mechanism (deliberately simple, no obvious kickers/ramps) |
| INDEXER / HOPPER<br>Spindexer 'bowl' with a central cone wrapped in cat-tongue grip tape to fling fuel outward to the feeder | DRIVETRAIN<br>Swerve (specifics unconfirmed)                                                         |
| ENDGAME<br>Climber not detailed.                                                                                             | AUTONOMOUS<br>Minimalist, low-motor spindexer; auto routines not detailed.                           |
| VISION<br>Unconfirmed                                                                                                        | PUBLIC CAD / DESIGN<br>No public CAD (reveal video)                                                  |

## LESSONS FOR STUDENTS

- Reputation for 'doing things with fewer motors than anyone thinks possible' — fewer actuators means fewer failure modes, lighter/cheaper robots.
- A grip-taped cone in a rotating bowl can index fuel with almost no dedicated hardware — elegant minimalism.

Video: <https://youtu.be/tOYFbRIJ8CA> · Reveal 'Enigma'

## REBUILD story: 'Roman I' (turret + low-compression spindexer, ~8 BPS) -&gt; 'Roman II', a WIDE aluminum DRUM shooter at ~23 BPS with a net hopper (~60-70 fuel). World-high single-match score [BtB transcript]

|                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SHOOTER<br>Roman II: wide aluminum DRUM (custom 3D-printed segments over an ordered tube, CatTongue-taped) + an adjustable hood; flywheel speed and hood angle scale with distance (further = faster + hood up); ~23 BPS (vs ~8 on the Roman I turret). A safety routine auto-lowers the hood entering the trench, with a bigger safety zone the faster it crosses [BtB transcript] | INTAKE<br>Custom silicone wheels (numbers printed) + an agitator linking all the rollers, CatTongue-taped for grip; a compact redesign over a wider v1 (a long bar that kept bending on hits was removed; flexible polycarb instead of Lexan to absorb impact); custom 3/8in-hex stub rollers instead of full-length shafts to save weight [BtB transcript] |
| INDEXER / HOPPER<br>A net hopper raised capacity from ~35 to ~60-70 fuel; rollers + agitator feed CatTongue-taped rollers up into the drum (the whole top net/cover flips up for easy access to the battery, PDH, and motors) [BtB transcript]                                                                                                                                      | DRIVETRAIN<br>Swerve; built for serviceability (the top flips up to reach the battery / PDH / intake & swerve motors); prototyped on a wooden chassis to place electronics before final build [BtB transcript]                                                                                                                                              |
| ENDGAME<br>Climber not detailed.                                                                                                                                                                                                                                                                                                                                                    | AUTONOMOUS<br>Vision-aimed turret; a safety routine auto-lowers the hood entering the trench (bigger zone the faster it crosses).                                                                                                                                                                                                                           |
| VISION<br>THREE Limelights tracking AprilTags (added a 3rd on the shooter over the previous two side cameras for more accurate pose); pose-based controls - one button locks onto the target / passing mode, and an operator button auto-aligns even if pose drops [BtB transcript]                                                                                                 | PUBLIC CAD / DESIGN<br>Public CAD — both versions (Onshape)                                                                                                                                                                                                                                                                                                 |

## LESSONS FOR STUDENTS

- Zero-compression indexing (move fuel by geometry/rotation, don't squeeze it) reduces jams and wear.
- A simple 'upkicker' (static bar + side wheels) is a low-part-count hand-off from spindexer to turret.
- Released TWO full CAD versions (Roman I & II) — a great study in in-season iteration.

Onshape CAD: <https://cad.onshape.com/documents/014716f840fdbd882c1d691f> · Reveal 'Roman' · Behind the Bumpers · Roman II CAD

## 'Big dumper' with a hopper-floor + drum shooter; simplicity-maximizing design. PNW District Champion, World Finalist [BtB transcript]

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                    |
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| <b>SHOOTER</b><br>4 lb, 4in-wide aluminum DRUM coated in CatTongue tape, ~3000 rpm; 24-25 BPS average (peaks 29-30), 4x Kraken X60; X44 hood with NO rollers (rollers hurt accuracy); hood -10deg to 60deg (60deg = bump-to-bump pass; -10deg shoots BACKWARD against the hub under defense while locking the wheels) [BtB transcript]                                                                                                                                                                                            | <b>INTAKE</b><br>Collector on 2x Kraken X60; two sets of CatTongue rollers + a 3rd 'pop-up' roller; stub axles for weight/durability; NO front beater bar - 1/2in polycarb + aluminum plates at contact points for resilience without the weight; an upper TPU-mounted protector roller keeps balls in the hopper [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>Hopper extends 11.5in (kept under 12in so flex doesn't breach frame perimeter), double-fold polycarb top for rigidity + a net for capacity; ~70-80 fuel full, 50-60 while still fitting under the trench. Hopper bed = 11 rollers at a 21:20 ratio (back roller faster for grip), belts ~0.03in loose to cut current; floor pivots up for battery access. S-shaped (vs J-shaped) superstructure with a single 2-Kraken feed roller for throughput + less jamming + more hopper volume [BtB transcript] | <b>DRIVETRAIN</b><br>Swerve (AdvantageKit-style logging of swerve states + odometry) [BtB transcript]                                                                                                                                                                                                                              |
| <b>ENDGAME</b><br>Climber not detailed (big dumper, simplicity-max).                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>AUTONOMOUS</b><br>Drive-and-empty dumper; auto routines not detailed.                                                                                                                                                                                                                                                           |
| <b>VISION</b><br>Vision pose updates fused with swerve odometry (logged); specific camera/coprocessor not named in the walkthrough (2025 code used PhotonVision + OV9782 AprilTag cameras) [BtB transcript]                                                                                                                                                                                                                                                                                                                       | <b>PUBLIC CAD / DESIGN</b><br>No confirmed 2026 release (GrabCAD history)                                                                                                                                                                                                                                                          |

### LESSONS FOR STUDENTS

- 'MkII' naming reflects a deliberate v1→v2 redesign — iterate on a known-good base rather than restarting.
- Known for fabrication quality and wire management — clean execution scales to deep playoff runs.
- Pre-champs EPA peaked ~301, one of the highest all season — a peak-performance benchmark.

**GrabCAD:** <https://grabcad.com/frc.2046.bear.metal-1> · TBA 2026 · [Behind the Bumpers](#)



### 'Cyclone': turret-mounted DYE-ROTOR shooter (up to 720deg tracked) fed by a four-bar full-length intake; shoot-on-the-move, ~18 fuel/sec. Heartland Regional [BtB transcript]

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Low-profile (4.25in tall) to clear the trench. 'Floating hood' on a cycloidal 80.5:1 + a 10 lb constant-force spring for rigidity; main flywheel = 4in WCP urethane 60A on a DEAD axle, 2x Kraken X60 (24:30 ratio), deliberately NO added flywheel mass so RPM changes instantly for shoot-on-move; separate 1in 45A backspin wheels on their own powertrain (1x X60); hood on a Kraken X44, up to 90deg; partial-depth pocketing for weight [BtB transcript] | <b>INTAKE</b><br>Four-bar FULL-LENGTH intake (holds up to 5 balls per intake); 5-roller system - the 5th roller adds velocity to push balls deep and clear dead zones; aluminum caps keep the polycarb rollers on their hubs through hits; a free-spinning 'Zamboni wheel' lets it run along the wall without being ripped off; motors hidden down low for packaging [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>DYE ROTOR (centering): a big outer diameter + a thin inner column; driven COAXIALLY (rotor speed independent of the wheel, tuned in code) via #25 chain (one motor wheels, one rotor); the rotor's metal rods carry much of the turret weight on two X-contact bearings; ~2.5 rot/sec. The 720deg turret runs #25 chain + a MAXPlanetary 25:1 with a 2:1-gear absolute encoder [BtB transcript]                                                       | <b>DRIVETRAIN</b><br>Swerve (its module velocities feed the shoot-on-the-move solver) [BtB transcript]                                                                                                                                                                                                                                                                                |
| <b>ENDGAME</b><br>Climber not detailed.                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>AUTONOMOUS</b><br>Turret dye-rotor (tracked up to 720deg) with shoot-on-the-move.                                                                                                                                                                                                                                                                                                  |
| <b>VISION</b><br>THREE Limelights for real-time pose. Shoot-on-the-move predicts a FUTURE position (swerve velocity x pose vector, scaled by the ~1s ball flight time) and aims there; the inertia-free dead-axle flywheel lets RPM jump (e.g. 50->20 RPS) instantly. Intake power auto-scales by drive direction to save current [BtB transcript]                                                                                                                               | <b>PUBLIC CAD / DESIGN</b><br>No public CAD (Behind the Bumpers)                                                                                                                                                                                                                                                                                                                      |

#### LESSONS FOR STUDENTS

- Turret wiring is hard: v1 over-built the umbilical; v2 ('C.U.R.S.E.D.' concealed umbilical retraction) routed wires down one protected conduit — 'less is more'.
- Minimal superstructure (one tube) keeps the fuel path to the rotor clear — let the game-piece flow path drive the frame.

**Video:** <https://www.youtube.com/watch?v=FbkE2kAgKQE> · [Reveal 'Cyclone'](#) · [Behind the Bumpers](#)

### Fixed full-width high-throughput floor-loader; volume shooter, NO turret

|                                                                                                                                                                                                    |                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Full-width hooded drum shooter, 4x Kraken X60; 1/16" aluminum drum surfaced with CatTongue tape; 3D-printed hood segments between 1/16" plates; separate Kraken X60 entry roller | <b>INTAKE</b><br>4-bar intake (roller plates, side hopper plates, upper links in SRPP); static kicker bars + 2x Kraken X60 rollers; deploy off a MaxSpline shaft under the floor |
| <b>INDEXER / HOPPER</b><br>Roller floor on WCP dead-axle stubs; belt shortened 0.030" to cut current; moving hopper sits under trench height; stretchy net gives ~65 fuel                          | <b>DRIVETRAIN</b><br>Swerve (modules unconfirmed)                                                                                                                                |
| <b>ENDGAME</b><br>Climber not detailed (fixed full-width, no turret).                                                                                                                              | <b>AUTONOMOUS</b><br>Volume shooter; auto routines not detailed.                                                                                                                 |
| <b>VISION</b><br>Unconfirmed                                                                                                                                                                       | <b>PUBLIC CAD / DESIGN</b><br>No public CAD (detailed reveal writeup); GitHub                                                                                                    |

#### LESSONS FOR STUDENTS

- You don't need a turret to be competitive — a full-width fixed shooter + huge hopper wins on throughput.
- SRPP and dead-axle construction keep a wide superstructure light and stiff.
- Tiny tuning (0.030" belt shortening) measurably reduced current draw — budget current deliberately.
- A compliant net hopper trades rigidity for capacity (~65 fuel).

**CD thread:** <https://www.chiefdelphi.com/t/team-1768-nashoba-robotics-2026-robot-reveal-nightshift/519386> · [Reveal 'Nightshift'](#) · [Behind the Bumpers](#)

#26

**8044 Denham Venom** Cat-5

Denham Springs, LA

**EPA 257**

50-9

**Ran two robots: v1 (low 'bump'/under-trench) and v2 (a HighTide/Denham 'tide clone'); dye-rotor shooter [per 8044 team members on DDS]**

|                                                                                                                                                                                                                    |                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Dye-rotor shooter, belt-driven (HighTide-style); v1 had belt skip only when jammed, v2 fixed it by tensioning the belt rather than adding 10DP gear stages [per 8044 on DDS - medium confidence] | <b>INTAKE</b><br>Unconfirmed (2025 robot noted for a clean intake-to-handoff path — likely carryover) |
| <b>INDEXER / HOPPER</b><br>Dye-rotor fed (rotor flings fuel to the launcher) [per DDS photos]                                                                                                                      | <b>DRIVETRAIN</b><br>Swerve (specifics unconfirmed)                                                   |
| <b>ENDGAME</b><br>Climber not detailed (ran v1 under-trench + v2).                                                                                                                                                 | <b>AUTONOMOUS</b><br>Auto routines not detailed.                                                      |
| <b>VISION</b><br>Unconfirmed                                                                                                                                                                                       | <b>PUBLIC CAD / DESIGN</b><br>No public CAD (DDS robot-pics photos)                                   |

**LESSONS FOR STUDENTS**

- A 50-9 record with little public documentation: execution + driver practice can outrank publicity.
- Their history of slick mechanism handoffs is worth studying for reliable ball transfer.

**DDS thread:** <https://discord.com/channels/1120162219502608426/1479491912305217637> · [DDS robot-pics-2026](#) · TBA · [DDS robot-pics-2026 thread \(8044\)](#)

#27

**1706 Ratchet Rockers** (undisclosed)

Wentzville, MO

**EPA 256**

45-2

**Dual-turret shooter ('Mirage' - 'shooting so fast you think you're seeing double') fed by a single spindexer; Oklahoma Regional winner [BtB transcript]**

|                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>TWO turrets, each with a front kicker + a back kicker; 6in aluminum flywheels (made by sponsor Custom Machine Works - tried steel, aluminum won); hoods rotate fully [BtB transcript]                                                        | <b>INTAKE</b><br>Full-width four-bar intake, 2x Kraken X60 (one for pitch, one for the rollers); 4 silicone rollers (3 carbon-fiber + 1 polycarbonate, all wrapped in white silicone) for grip; the intake doubles as a hopper extension to minimize moving parts [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>A spindexer (re-iterated several times over the season) ramps fuel up into each shooter; kicker wheels fire the front + back kicker on each turret [BtB transcript]                                                                 | <b>DRIVETRAIN</b><br>Swerve (odometry fused with vision in the pose estimator) [BtB transcript]                                                                                                                                                                                    |
| <b>ENDGAME</b><br>Climber not detailed.                                                                                                                                                                                                                        | <b>AUTONOMOUS</b><br>Dual-turret very high fire-rate ('Mirage'); auto routines not detailed.                                                                                                                                                                                       |
| <b>VISION</b><br>3x Limelight 4 + swerve odometry in the pose estimator; the Limelight hardware-manager Kalman filter rejects extraneous poses and only trusts MegaTag2 on good solutions; Limelights sit under each shooter + on the climber [BtB transcript] | <b>PUBLIC CAD / DESIGN</b><br>No public CAD; code on GitHub                                                                                                                                                                                                                        |

**LESSONS FOR STUDENTS**

- A 45-2 record is a case study in consistency over flash — minimize failure modes, maximize cycle reliability.
- Watch their Behind the Bumpers for how a veteran team packages a clean, serviceable robot.

**Code (GitHub):** <https://github.com/rr1706> · [Behind the Bumpers](#) · TBA 2026

## 2-wide drum 'over-bumper reverse' shooter (modeled on 254's 2017 'Misfire')

|                                                                                                                                                                                                   |                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>2-wide drum shooter, fixed 34deg hood (polycarb + foam, ~0.675in compression); 4 motors on one shaft (SDS brass + TTB-0106 flywheels, ~5.2 lb); flywheel RPM scaled by distance | <b>INTAKE</b><br>Roller floor with 'pulsing' (reverse-intake while shooting to empty the hopper) |
| <b>INDEXER / HOPPER</b><br>Roller floor: indexer rollers 0.375in, top floor roller 0.5in; indexer/floor/shooter run 1:1 at 12V                                                                    | <b>DRIVETRAIN</b><br>MK5N-style swerve, ~26x26in, Spike grip wheels                              |
| <b>ENDGAME</b><br>Climber not detailed.                                                                                                                                                           | <b>AUTONOMOUS</b><br>2-wide drum 'over-bumper reverse' shooter; auto routines not detailed.      |
| <b>VISION</b><br>Single AprilTag camera for pose (primarily off the hub tags); a 2nd camera under consideration                                                                                   | <b>PUBLIC CAD / DESIGN</b><br>Open Alliance build log + GitHub                                   |

### LESSONS FOR STUDENTS

- A young team (3rd year) reaching #28 EPA shows the value of an Open Alliance build log — public iteration accelerates learning.
- Out-of-state district scheduling (VA→GA→SC) is a program-building lesson, not just a robot one.

**CD thread:** <https://www.chiefdelphi.com/t/lynk-9496-build-log-2026/511797> · [Open Alliance build log](#) · [Reveal 'Matterhorn'](#)

## Drum shooter mounted on a turret (a 'turret dumper'), ~30deg total turret travel

|                                                                          |                                                                                 |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Drum shooter on a ~30deg-travel turret                 | <b>INTAKE</b><br>Unconfirmed                                                    |
| <b>INDEXER / HOPPER</b><br>Unconfirmed                                   | <b>DRIVETRAIN</b><br>Swerve (historically SDS MK4i)                             |
| <b>ENDGAME</b><br>Climber not detailed (drum-on-turret 'turret dumper'). | <b>AUTONOMOUS</b><br>Auto routines not detailed.                                |
| <b>VISION</b><br>Unconfirmed                                             | <b>PUBLIC CAD / DESIGN</b><br>2026 release pending (strong tech-binder history) |

### LESSONS FOR STUDENTS

- Strike Zone habitually publishes a full tech binder (2023, 2025) — review those as a documentation template.
- A team known for 'accurate' shooters underscores investing in shot consistency over raw range.

**CD thread:** <https://www.chiefdelphi.com/t/5460-strike-zone-2026-reveal/515689> · [2026 'Helios' reveal](#) · [2025 CAD + binder](#)

**'Rubble': dye-rotor turret shooter w/ 35-deg adjustable hood, pivoting floor intake, WCP X2i/X2c swerve**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Dye-rotor turret ('floating turret' de-conflicts rotor table vs turret). Drum: 4 in OD x 1/16 in-wall aluminum wrapped in Cat Tongue grip tape, powered by four Kraken X60s. Adjustable hood with 35 deg of rotation on a Kraken X44 at 1:120 through a zombie roller; two 2 in Cat Tongue hood rollers geared up 1.64:1 plus three 1.5 in silicone-sleeved rollers on two X60s (1:1.5). Chose the short dye-rotor turret over a large dumper after week-1 Alpha Bot testing hit ~15 BPS but lacked consistency. | <b>INTAKE</b><br>Pivoting floor intake: pivots up/down for bellypan access and ball compaction; elastic X-pattern springing keeps constant tension so balls don't jam under the floor, and the front roller is lowered so balls can't cam under during the compaction sequence. Custom Onyx-printed pulleys with 1/16 in steel dowels to prevent layer shearing; installs onto the shooter tower with two bearing plates. |
| <b>INDEXER / HOPPER</b><br>Floor rollers feed the shooter tower. A CANrange measures hopper fill and auto-runs the rollers until a retroreflective sensor in the tower trips - buffering ~8 extra fuel in the tower without ejecting prematurely.                                                                                                                                                                                                                                                                                  | <b>DRIVETRAIN</b><br>28.5 x 27 in frame (1/8 in-wall tube, 5/8 in corner chamfers to stay under the 110 in perimeter). Two WCP X2i + two X2c swerve modules, X1 gearing with 10T pinions, Kraken X60 drive / X44 steer; custom 1/16 in bottom plates to drive over the Depot, intake-side module top plates cut down for a lower intake.                                                                                  |
| <b>ENDGAME</b><br>Climber not detailed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>AUTONOMOUS</b><br>Dye-rotor turret with adjustable hood (chosen over a dumper after Alpha-Bot testing); auto routines not detailed.                                                                                                                                                                                                                                                                                    |
| <b>VISION</b><br>3x Limelight 4 (rear 'SHOOTER' + left + right), AprilTag MegaTag2; Limelight-only pipeline (no PhotonVision)                                                                                                                                                                                                                                                                                                                                                                                                      | <b>PUBLIC CAD / DESIGN</b><br>Full robot CAD (Onshape) + code (GitHub, team581/frc-2026)                                                                                                                                                                                                                                                                                                                                  |

**LESSONS FOR STUDENTS**

- Week-1 'Alpha Bot' archetype test: they built a quick dumper, measured ~15 BPS but poor consistency, and let DATA kill the archetype before committing - cheap insurance against a season-long mistake.
- Software power manager: a separate thread re-allocates current limits across subsystems in real time (e.g. prioritize the shooter while scoring) to prevent brownouts over a full match.
- Defensive autonomy: if all cameras drop, the robot tells the drivers and falls back to pre-set shots; dashboard toggles disable complex features mid-match.
- Turret wire management without e-chain: cable sleeving + surgical tubing for tension - low-cost and light.

**Onshape CAD:** <https://cad.onshape.com/documents/766d37b113335d7a60cc2375/w/8b35fc383279c1954cec5b72/e/bd3188f47c23702bef9f1fc8> · CAD + Code release (CD) · Reveal (full Q&A) · Code (GitHub)

## Additional well-documented robots for study (beyond the EPA top-30)

These rank below the EPA top-30 but each has a strong public release (CAD, code, binder, or detailed reveal) and a clear design lesson. Ordered by EPA. ★ marks the recommended study picks.

|                  |                                                  |                  |
|------------------|--------------------------------------------------|------------------|
| #34 (study pick) | <b>5000 Hammerheads</b> Icarus<br>Marblehead, MA | EPA 236<br>41-23 |
|------------------|--------------------------------------------------|------------------|

### Turreted vision-tracking fuel scorer; full Open Alliance + open-source code

|                                                                                                                            |                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Turret-mounted shooter (prototype → SRPP V2 ~85 lb); camera-tuned auto-aim                               | <b>INTAKE</b><br>Rack-and-pinion deploy (TalonFX rack motors, ~48:17 & 50:17 belt reductions, 0.5in pinion) + spin rollers |
| <b>INDEXER / HOPPER</b><br>Spin motor + dual feed motors (feed + follower) with unjam logic and hook/feed stall detection  | <b>DRIVETRAIN</b><br>Swerve (CTRE/TalonFX)                                                                                 |
| <b>ENDGAME</b><br>Climber not detailed.                                                                                    | <b>AUTONOMOUS</b><br>Turreted vision-tracking auto-aim; full Open Alliance open-source code.                               |
| <b>VISION</b><br>6x Arducam OV9281 on PhotonVision (front L/R/center + back L/R/center), MegaTag, 2026 welded-field layout | <b>PUBLIC CAD / DESIGN</b><br>Full public CAD (Onshape) + open code                                                        |

#### LESSONS FOR STUDENTS

- Live Onshape CAD + GitHub code (incl. a TurretIoTalonFX subsystem) is a goldmine for studying turret control architecture.
- Iterating a turret from prototype → SRPP V2 shows real weight-reduction practice.
- PhotonVision turret tracking tuned for both aim speed and defense is a concrete vision-integration case study.

**Onshape CAD:** <https://cad.onshape.com/documents/a347ccba68f245233aadb8b> · [Open Alliance thread \(design post\)](#) · [GitHub](#) · [Onshape](#)

## Turret + adjustable-hood flywheel shooter, slapdown intake, two-stage tunnel+accelerator indexer, sliding hopper, swerve, climber; V1 & V2 CAD + 28-page binder

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Adjustable-hood flywheel on a turret. Flywheel: 2x Kraken X60 (1:1) spinning three 4" 60A Stealth wheels + two trimmed SDS 4" brass flywheels (added moment of inertia), with five 1" Sushi accelerator rollers and a 3D-printed back ramp to compress the FUEL. Hood: Kraken X44 (167.5:1), 3D-printed sector-and-pinion, 28 deg range. Turret: 7.5" ID X-Contact bearing, custom 8ST/10DP gear, +/-200 deg (400 deg total) on a Kraken X44 (50:1), biased to the back-left corner for shoot-on-the-move.          | <b>INTAKE</b><br>Motor-powered slapdown, three full-width horizontal rollers, touch-it-own-it. Deploy: Kraken X44 (50:1) via #25 chain on a coaxial shaft (also drives the floor belts) plus a spring-loaded kicker bar; the deployment motion also actuates the prismatic sliding hopper. Rollers: Kraken X60 (1.60:1), 1.5" polycarb tubes in silicone; 1/4" pocketed 7075 side plates, SRPP front guard. (V2 rebuild swapped this slapdown for a linear intake whose plates directly carry the extending hopper.) |
| <b>INDEXER / HOPPER</b><br>Two stages. Tunnel (Kraken X60, 1.88:1): top + bottom conveyors, two 15 mm belts each, align the FUEL into one controlled stream. Accelerator (Kraken X44, 1.46:1): three 2" Stealth + custom-lathed 2.5" Colson wheels fling the FUEL up into the turret. Fed by a sliding/extending hopper and a floor conveyor (back kicker roller -> Kraken X60, 6.30:1).                                                                                                                                              | <b>DRIVETRAIN</b><br>SDS Mk5n R3 swerve (4 modules), Kraken X44 drive + steer, 26.09:1 steering, 4" molded spike-grip wheels (2.25" wide), 19.2 ft/s. 28x26" pocketed aluminum frame; Pigeon 2.0 IMU + CTRE CANcoders on all four modules.                                                                                                                                                                                                                                                                           |
| <b>ENDGAME</b><br>Has a climber (per the 28-page binder); part of the V1->V2 design.                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>AUTONOMOUS</b><br>Single-driver, two-button automation: auto-aim + auto spin-up by field position, hood auto-lowers under the trench; PathPlanner autos; 5-camera AprilTag localization; shoot-on-the-move.                                                                                                                                                                                                                                                                                                       |
| <b>VISION</b><br>Five cameras feeding full-field AprilTag localization (robot pose, vision targets, and trajectories all modeled in physics simulation), logged via AdvantageKit / AdvantageScope 3D replay; specific camera brand not named in the binder. Drives single-driver, two-button stateless automation: auto spin-up + auto-aim to the hub or an alliance-zone corner by field position, with the hood auto-lowering under the trench; PathPlanner autos with a profiled point-to-point PID controller for precise climbs. | <b>PUBLIC CAD / DESIGN</b><br>CAD for both robots (Onshape) + 28-page technical binder                                                                                                                                                                                                                                                                                                                                                                                                                               |

### LESSONS FOR STUDENTS

- A two-stage indexer — a 'tunnel' that single-streams the FUEL, then an 'accelerator' that adds exit velocity — cleanly decouples organizing balls from launching them.
- Stateless, two-button automation (one intake, one shoot) let them run a SINGLE driver: the robot auto-aims and auto-spins by field position and auto-lowers the hood under the trench.
- A mid-season rebuild (V1 slapdown + rail-carriage hopper -> V2 linear intake + directly-coupled hopper) is packaging-driven iteration; both versions' CAD are public for an A/B study.
- Validating every subsystem in physics-based simulation (FUEL, vision targets, and trajectories modeled) sped iteration and de-risked hardware before the robot existed.

**CAD release:** <https://www.chiefdelphi.com/t/team-359-hawaiian-kids-2026-cad-release-q-a/521575> · [CD CAD release + Q&A](#) · [Statbotics](#) · [The Blue Alliance](#)

### Ultra-clean sheet-metal turret shooter; #1 EPA in China, championship build quality

|                                                                                                                                 |                                                          |
|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>SHOOTER</b><br>Custom CNC-machined flywheel (bore geometry designed to be cut with a 2.5mm end mill); turret-mounted shooter | <b>INTAKE</b><br>Roller intake (motors unconfirmed)      |
| <b>INDEXER / HOPPER</b><br>Indexer/hopper present (capacity not stated in thread)                                               | <b>DRIVETRAIN</b><br>Swerve (module unconfirmed)         |
| <b>ENDGAME</b><br>Climber not detailed in the thread.                                                                           | <b>AUTONOMOUS</b><br>Auto routines not detailed.         |
| <b>VISION</b><br>Unconfirmed (not detailed in CAD thread)                                                                       | <b>PUBLIC CAD / DESIGN</b><br>Full CAD release (Onshape) |

#### LESSONS FOR STUDENTS

- Design parts AROUND your tools (DFM): they shaped the flywheel bore so a 2.5mm end mill could cut it.
- Outsource sheet metal to a sponsor, machine the rest in-house - that division of labor builds a clean, precise robot fast.
- Tidy wire management and packaging correlate with reliability and top EPA - visual order is an engineering signal.

**CAD release:** <https://www.chiefdelphi.com/t/frc-cyber-unicorn-8214-2026-champ-robot-cad-release/521310> · [CD champ-robot CAD release](#) · [Statbotics](#)

#56 (study  
pick)

**4481 Team Rembrandts** (Open Alliance)

Eindhoven, NL

**EPA 214**

37-15

### Precision turreted hooded-flywheel scorer; strong controls

|                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Turret hooded flywheel - their 4TH shooter design: a static back wall (they tried twin rollers but found them unneeded for high BPS), Teflon tape to cut friction/backspin, a custom flywheel to hold RPM, and a switch to METAL gears (the 3D-printed ones damaged easily) for champs. (Underlying: dual NEO Vortex single axle, ~370deg travel, herringbone rack-pinion hood 15-45deg) [BtB transcript] | <b>INTAKE</b><br>PASSIVE LINEAR intake (slides in/out, doesn't fold): NO deploy motors - constant-force springs + gravity drop it at match start; can't retract mid-match (deemed unnecessary); very impact-resistant since it can fold in on a front hit without breaking [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>'Washing machine' spindexer: spins very fast; iterated to a LARGER outer chamber + inner cone (helps balls settle -> higher BPS) with the gearbox + two motors packaged INSIDE the cone; lowered the washing-machine walls to admit balls easier. Turret cable chain kept taut by an elastic band (shortest at the anchor, stretches as it spins - never snags) [BtB transcript]                 | <b>DRIVETRAIN</b><br>Swerve (not detailed in the walkthrough)                                                                                                                                                                                                                               |
| <b>ENDGAME</b><br>Climber not detailed.                                                                                                                                                                                                                                                                                                                                                                                     | <b>AUTONOMOUS</b><br>Precision turreted hooded-flywheel with vision auto-aim; strong controls.                                                                                                                                                                                              |
| <b>VISION</b><br>Strong vision system: FOUR cameras at different angles to approximate robot pose; heavy focus on power management (per-subsystem current limits + supply voltage) [BtB transcript]                                                                                                                                                                                                                         | <b>PUBLIC CAD / DESIGN</b><br>Public CAD (GrabCAD) + Open Alliance thread                                                                                                                                                                                                                   |

#### LESSONS FOR STUDENTS

- Excellent gear-ratio documentation (45.7:1 flywheel, 58.5:1 hood) — a model for sizing reductions to control-vs-speed needs.
- A through-bore encoder on the hood enables true closed-loop angle control, not open-loop presets.
- Their turret cable-sleeve fix (lengthening it to go from ~270° to ~370°) — design the wire path for full range from the start.
- Authored the widely-used '3DM' game-analysis document.

**GrabCAD:** <https://grabcad.com/team.rembrandts.4481-1> · [Open Alliance thread](#) · [GrabCAD](#) · [Behind the Bumpers \(full transcript\)](#)

**'Sandspit': four-bar intake -> pleated-belt roller floor -> full-width DRUM shooter (NO turret). Flow-obsessed; ~15 BPS. Orange County District [BtB transcript]**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
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| <b>SHOOTER</b><br>Full-width DRUM shooter (~15 BPS) with two massive ~5in in-house-machined flywheels (steel compound, mass concentrated at the outer radius for max inertia-per-weight, cut on a Haas mini-mill over a multi-day, multi-tool process); active hood on an in-house 10DP gear, symmetrical with a shaft + a 1:1 through-bore encoder for shoot-and-move; 3D-printed hood backings [BtB transcript]                                                                                 | <b>INTAKE</b><br>Four-bar intake with active ball control (its linear push turns the intake into a ramp funneling fuel toward the shooter); a 'scotch yoke' makes the hopper box follow the intake up/down; custom-machined 6061 sprockets (stock shortage); pivot driven through a 1in round tube to both ends. Tried coaxial roller power (balls caught) -> switched to Citrus's double-motor intake (much better) [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>A 2:1 gearbox drives the roller floor + the first feeder roller; 'pleated belts' (TPU-printed cleats) on 3D-printed CF crown rollers agitate + push balls linearly down the ramp (replacing two top rollers that let balls fall back); crown belts drive the whole floor without belts between every roller (less friction). An 'epoxy coupler' roller mount (epoxy + embedded 4-40 nuts) cuts ~50% of stub-roller weight + allows quick roller swaps [BtB transcript] | <b>DRIVETRAIN</b><br>Swerve (not detailed publicly)                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>ENDGAME</b><br>Climber not detailed (focus is on ball flow).                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>AUTONOMOUS</b><br>Auto routines not detailed.                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>VISION</b><br>Switched from Limelight to PhotonVision: SIX cameras - four AprilTag (two forward, two back) + two rear OBJECT-DETECTION cameras that find the densest fuel; 1600x800 (sharper than Limelight) with a custom ML algorithm that crops each frame to the AprilTags so it computes only on them (keeps ~50 fps + a very stable pose); two Rubik Pi 3 co-processors [BtB transcript]                                                                                                 | <b>PUBLIC CAD / DESIGN</b><br>Reveal video only (no downloadable CAD)                                                                                                                                                                                                                                                                                                                                                                 |

#### LESSONS FOR STUDENTS

- Treat ball flow as THE design objective - anything a ball can sit on, wedge above, or spin on is the enemy.
- Dividers/lanes that look organized often trap balls; smooth continuous paths beat segmented ones.
- Make the feeder transparent and film with high-speed cameras to diagnose flow problems you can't see at speed.
- Use the intake/box as a compactor to push fuel into the feeder while shooting.

**CD thread:** <https://www.chiefdelphi.com/t/team-3476-code-orange-2026-sandspit-robot-reveal/516875> · [CD reveal + flow philosophy](#) · [Reveal video](#) · [Behind the Bumpers](#)



## Over-bumper slapdown intake -> moving floor -> uptake -> DRUM shooter (swapped from a 4-lane shooter mid-season); custom ROS2 software. New England DCMF [BtB transcript]

|                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>SHOOTER</b><br>4in TITANIUM drum + a 1:1 BRASS flywheel + two carbon-fiber back rollers at 2:1; ~25 BPS. Started as a 4-lane shooter (~15 BPS) but balls jammed on the lanes -> removed the lanes and went to the drum [BtB transcript]                                                                                                     | <b>INTAKE</b><br>Over-bumper SLAPDOWN deployed off an X60 (planetary 9:1 + two 3:1 for a very fast deploy, esp. in auto); 3 polycarb rollers - a bottom kicker (no silicone; it caught on the bump) + two silicone front rollers; one Kraken X60 ~3:1. Swapped from an earlier rack-and-pinion + 4-roller intake that stuck/broke on hits (new one lighter + more robust) [BtB transcript] |
| <b>INDEXER / HOPPER</b><br>Moving floor = WCP stub-roller system (carbon-fiber rollers with silicone) on one Kraken X60 (~1:1, pushes balls hard into the uptake); uptake = six 1in carbon-fiber silicone tubes on 2x Kraken X60 [BtB transcript]                                                                                              | <b>DRIVETRAIN</b><br>Swerve (auto-shoot puts the swerve in X-mode to resist defense) [BtB transcript]                                                                                                                                                                                                                                                                                      |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                                                                                                                                                                                                                                                                         | <b>AUTONOMOUS</b><br>Custom CyberKraken power/CAN boards; auto routines not detailed.                                                                                                                                                                                                                                                                                                      |
| <b>VISION</b><br>FOUR Arducams -> a Mac Mini running PhotonVision; if any camera sees 2 AprilTags it fuses with odometry for full field awareness. Runs ROS 2 (a node per subsystem, so one crash doesn't take down the robot) + Molex locking connectors; full match logging (all node messages + 4 camera feeds) for replay [BtB transcript] | <b>PUBLIC CAD / DESIGN</b><br>Full CAD + code + docs (Onshape + SolidWorks)                                                                                                                                                                                                                                                                                                                |

### LESSONS FOR STUDENTS

- Diagnose and fix recurring failures: they cured belt-snap by swapping to chain at the highest-load point.
- Custom 'CyberKraken' power/CAN boards with Anderson connectors (chosen over Molex SL for shock resistance) - real electrical-design depth.
- Win by refining a proven architecture through continuous in-season optimization.
- Public ROS-based robot code is an advanced software study target.

**CAD release:** <https://www.chiefdelphi.com/t/team-195-the-cyberknights-2026-cad-code-and-documentation-release/521390> · [CD CAD/code/docs release](#) · [Onshape CAD](#) · [Behind the Bumpers](#) · [GitHub](#)

## Software-first team: reliable turret shooter built around AdvantageKit + Northstar vision

|                                                                                                                                                                                                                                        |                                                                                                                                                                         |
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| <b>SHOOTER</b><br>Turret shooter with adjustable 3D-printed (PLA+) hood; 4in WCP Fairlane main wheel + 2in solid-core flex top wheel on a live single shaft; optional brass flywheel for added inertia; hood gear routed from WCP SRPP | <b>INTAKE</b><br>Compliant-wheel roller intake (prototyped publicly) feeding a centering/serializing path                                                               |
| <b>INDEXER / HOPPER</b><br>Intake -> centering hopper -> tower uptake -> turret/flywheel; serialization handled in software so balls feed one at a time                                                                                | <b>DRIVETRAIN</b><br>Swerve; .090/.063 5052 aluminum sheet with lightening tricks                                                                                       |
| <b>ENDGAME</b><br>Climber not detailed in the build thread.                                                                                                                                                                            | <b>AUTONOMOUS</b><br>Software-first: AdvantageKit + Northstar AprilTag localization (Orange Pi); strong auto-aim / autonomous pedigree (routines in the released code). |
| <b>VISION</b><br>'Northstar' custom AprilTag pipeline on Orange Pi coprocessors; alpha camera mounted behind the shooter hood for turret-to-hub alignment; full AdvantageKit/AdvantageScope logging + sim                              | <b>PUBLIC CAD / DESIGN</b><br>Open + code (AdvantageKit); CAD links in build thread                                                                                     |

### LESSONS FOR STUDENTS

- AdvantageKit IO-layer architecture and deterministic input->output ordering - a gold-standard software study.
- Northstar AprilTag localization on Orange Pi is a buildable, open-source vision stack.
- Routing custom gear teeth (SRPP/10DP) for turret and hood - a manufacturing technique they openly teach.
- Live-axle shooter packaging and brass-flywheel inertia trade-offs.

**Code (GitHub):** <https://github.com/Mechanical-Advantage/RobotCode2026Public> · [CD build thread](#) · [Reveal 'Darwin'](#) · [GitHub \(public code\)](#)

**'Tempest': high-capacity swerve turret shooter — intake → hopper → spindexer → feeder →  $\pm 250^\circ$  turret → adjustable-hood flywheel, plus a climber**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Turret-mounted adjustable-hood flywheel. Hood 15–55° via rack (1× NEO 2.0, 15.37:1 MAX Planetary). Flywheel: 2× Kraken X60 (1.4:1) driving 2× brass inertia flywheels (~150 g ea) + 2× 4 in urethane (WCP 60A). Turret: 6.5 in X-contact bearing, 42.24:1 (96T/10T), 1× Kraken X44, $\pm 250^\circ$ . Shoot-on-the-move via calibrated interpolation maps (distance → hood angle / RPM / flight time), 20 iterations per cycle. | <b>INTAKE</b><br>'Touch-it-own-it' ground intake. 8 mm 7075 towers + 1/2 in polycarb arms (10DP rack); front 2.25 in (AM 35A) + rear 1.25 in rollers driven by 2× NEO Vortex (2:1, ~33 ft/s). Rack expansion by 1× NEO Vortex (5:1) deploys in under 1 s; 45° polycarb ramp. |
| <b>INDEXER / HOPPER</b><br>Hopper → Spindexer → Feeder. Hopper: polycarbonate + carbon-fiber, expands ~12 in to hold up to ~55 FUELS. Spindexer: 3D-printed cone, 2× NEO Vortex (4:1), keeps pieces moving (anti-jam). Feeder: 1× Kraken X60 (2.2:1), 16 compliant rollers, multi-piece flow with no dead zones.                                                                                                                                  | <b>DRIVETRAIN</b><br>Custom SDS MK4n swerve on a 28×26 in frame (2×1 in Al tube, 2 mm bellypan). L1+ (14.2 ft/s); Kraken X60 drive + Kraken X44 steer (field-oriented); machined hubs with TPU treads for grip.                                                              |
| <b>ENDGAME</b><br>Climber on the turret shooter (binder-documented).                                                                                                                                                                                                                                                                                                                                                                              | <b>AUTONOMOUS</b><br>Shoot-on-the-move via calibrated interpolation maps (distance->hood/RPM/flight-time), 20 iterations per cycle; 4-camera AprilTag fusion.                                                                                                                |
| <b>VISION</b><br>4-camera fusion: 2× Limelight + 2× Arducam → MultiCameraPoseEstimation with AprilTag corrections feeding odometry. Turret angle solved by trig + Chinese Remainder Theorem on two absolute encoders for a 720° usable range.                                                                                                                                                                                                     | <b>PUBLIC CAD / DESIGN</b><br>Full CAD (Onshape) + 32-page technical binder                                                                                                                                                                                                  |

#### LESSONS FOR STUDENTS

- A 3DM (data-driven) process drove the design: priority list + archetype decision matrix + functional requirements (e.g. <22.25 in to clear the trench, <21 kg modules for air travel).
- The binder documents real motor / ratio / material callouts for every subsystem — a model for a complete technical binder.
- Chinese Remainder Theorem on two absolute encoders yields a 720° turret range without re-zeroing — an elegant controls trick.
- Shoot-on-the-move uses experimentally-calibrated interpolation maps (not real-time physics) iterated 20×/cycle — a practical approach to moving shots.

**Onshape CAD:** <https://cad.onshape.com/documents/96bff94aa06158572d9954cd/w/156a28a4c439f1f7e5f3bf04/e/3a4d945dab8b9e666f1c061c> · [CD CAD + tech binder release](#) · [32-page technical binder \(Drive\)](#) · Statbotics

## Turret-on-hood shooter fed by a spindexer; physics-model firing solutions

|                                                                                                                                                                                                                                                                            |                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Turret-mounted flywheel + adjustable hood; firing solution (wheel speed, turret/hood angle) from a physics model (air resistance, Magnus force) instead of a lookup table; auto-opens hood after trench from pose/velocity; Kraken X44; SOTM in progress | <b>INTAKE</b><br>Floor intake on revised roller geometry tuned for rapid multi-ball pickup without jamming; used for center-field steals in auto                                 |
| <b>INDEXER / HOPPER</b><br>Spindexer -> curved ramp -> gate/kicker wheels; HDPE hopper walls (replaced polycarb) to reduce sticking; counter-rotating mecanum agitator (paintball-hopper inspired) forces balls down; 2x CANrange sensors; ~55-ball goal                   | <b>DRIVETRAIN</b><br>WCP SwerveX2c; Pigeon 2.0 IMU; PathPlanner autos for neutral-zone steals                                                                                    |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                                                                                                                                                                                                     | <b>AUTONOMOUS</b><br>Physics-model firing solution (wheel speed + turret/hood angle from an air-resistance & Magnus model); hood auto-opens after the trench from pose/velocity. |
| <b>VISION</b><br>PhotonVision on Orange Pi with AprilTags around the central target; local 3D pose algorithm                                                                                                                                                               | <b>PUBLIC CAD / DESIGN</b><br>Full Onshape CAD (Open Alliance)                                                                                                                   |

## LESSONS FOR STUDENTS

- A physics-based firing model beats a lookup table - re-tuning means changing a few parameters, not re-populating a table.
- Material choice fixes jams: swapping polycarb hopper walls to slipperier HDPE reduced sticking.
- Borrow cross-domain ideas - paintball counter-rotating agitators inspired their mecanum de-jam wheel.
- Program in simulation first so new students can write/tune code before the robot exists.

**CD thread:** <https://www.chiefdelphi.com/t/frc-5987-galaxia-2026-build-thread-open-alliance/505340> · [CD build thread](#) · [In-season subsystem post](#) · [GitHub](#)

## Strategy-first: wide touch-it floor intake + flow-through hopper + turret single-file shooter

|                                                                                                                                                                                                                                                             |                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Turret-mounted flywheel, single-file feed; ~8 balls/sec peak for ~2s then ~5 sustained; re-gear flywheel pulleys (15:30 -> 22:24) for higher speed; used motion-profile / current control (6328-inspired) to tame large startup PID error | <b>INTAKE</b><br>Wide touch-it-own-it floor intake (3 balls wide); ran ~50% speed to avoid ejecting balls over the hopper; also acts as a compactor |
| <b>INDEXER / HOPPER</b><br>Flow-through hopper (~45+ balls) with spindexer-style feed; throughput drops as far-from-spindexer balls settle slowly; adding agitators                                                                                         | <b>DRIVETRAIN</b><br>Swerve; emphasis on under-trench driveability + fast cycles vs double-wide shooters                                            |
| <b>ENDGAME</b><br>No climber - their strategy analysis flagged climbing as low-value.                                                                                                                                                                       | <b>AUTONOMOUS</b><br>Strategy-first; auto routines not detailed.                                                                                    |
| <b>VISION</b><br>Unconfirmed (AprilTag targeting expected; not detailed)                                                                                                                                                                                    | <b>PUBLIC CAD / DESIGN</b><br>Onshape CX26 assembly (Open Alliance)                                                                                 |

## LESSONS FOR STUDENTS

- Build an Alpha bot to surface integration problems early (melting sidewalls, chain backlash, hopper dead zones) before the comp design.
- Know your shooter's real throughput CURVE - peak BPS means little if it collapses once near-spindexer balls are gone.
- Flywheel PID from a dead stop oscillates; a motion profile / constant-current model tames startup error.
- Single-wide shooters lose a measured throughput race to double-wide; compensate with turret + driveability + faster cycles.

**CD thread:** <https://www.chiefdelphi.com/t/ceit-x-5406-2026-build-thread-open-alliance/511200> · [CD build thread](#) · [Alpha-bot throughput analysis](#) · [GitHub](#)

**'Mixtape': double-turret flywheel shooter (two shooters + spindexer) w/ lead-screw adjustable hoods; swerve**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                               |
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| <b>SHOOTER</b><br>DOUBLE turret - two flywheel shooters, each with an adjustable hood driven by a lead screw. The lead screw was chosen to minimize the turret's swept radius: it tucks into the empty 'sides' instead of making the shooter longer. One iteration let the screw creep upward and lock the hood down; fixed by tapping and bolting the bottom of the screw (fix not shown in CAD). Wheel/motor specifics: read from the Onshape model. | <b>INTAKE</b><br>Ground intake feeding the hopper/spindexer (specifics not stated in the CD thread - see CAD).                                |
| <b>INDEXER / HOPPER</b><br>Simple spindexer feeding both shooters. Team reasoning: at high BPS it is easier to index into TWO shooters than one; they considered reviving their famous 2017 dye rotor (similar ball-shooting game) but judged it too finicky after losing the mentor talent behind the 2017 design.                                                                                                                                    | <b>DRIVETRAIN</b><br>Swerve (module and gearing not stated in the thread - see CAD).                                                          |
| <b>ENDGAME</b><br>Climber not detailed (first public CAD year).                                                                                                                                                                                                                                                                                                                                                                                        | <b>AUTONOMOUS</b><br>Double-turret architecture; 2026 code in Java (a break from the historic C++ stack); auto routines in the released code. |
| <b>VISION</b><br>Custom in-house vision pipeline (971 tradition); 2026 hardware unconfirmed.                                                                                                                                                                                                                                                                                                                                                           | <b>PUBLIC CAD / DESIGN</b><br>Full robot CAD (Onshape) + code (GitHub, frc971/971-second-robot-2026)                                          |

**LESSONS FOR STUDENTS**

- A double turret converts a hard indexing problem into an easier one: two simple feed paths at moderate rate instead of one path at very high rate.
- Lead-screw hood actuation as a packaging move - use the dead space beside the turret to shrink swept radius rather than growing the shooter envelope.
- Institutional knowledge is a real design constraint: they rejected their own well-known 2017 dye rotor because the people who understood it were gone.
- First year of public 971 CAD, and 2026 code in Java (a break from their historic in-house C++ stack) - a newly study-able, established program.

**Onshape CAD:** <https://frc971.onshape.com/documents/cabaa0c1c77517916df80783/w/48cb057db38a03cc202ff44e/e/96844befd4591dac162d657b> · CAD + Code release (CD) · CAD (Onshape) · Code (GitHub) · Statbotics

**Hopperless 360+ turret shooter with a 4-ball-wide intake and a direct 4-2-1 ball path**

|                                                                                                                                                                 |                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Flywheel + fully adjustable hood (team's first) on a 360+ degree turret; fed straight from intake to shooter (no storage hopper)              | <b>INTAKE</b><br>4-ball-wide bottom roller intake                                                                            |
| <b>INDEXER / HOPPER</b><br>No hopper - a '4-2-1' path narrows 4 lanes to 2 to 1 directly into the turret; ~8-ball working capacity                              | <b>DRIVETRAIN</b><br>WCP X2i swerve, 27in square base, 115 lb                                                                |
| <b>ENDGAME</b><br>Lightweight L1 climber kept deliberately cheap so mass/time goes to scoring.                                                                  | <b>AUTONOMOUS</b><br>360+ turret + adjustable hood for auto-aim; the hopperless 4-2-1 feed supports a fast continuous cycle. |
| <b>VISION</b><br>Single turret-mounted Limelight ('limelight-turret'), MegaTag2, IMU-fused; PhotonVision IO also present; ~8.25in turret-relative camera offset | <b>PUBLIC CAD / DESIGN</b><br>Full CAD + code + binder + pre-match checklists                                                |

**LESSONS FOR STUDENTS**

- Going 'hopperless' with a continuous 4-2-1 narrowing path removes a storage subsystem and its jam points - viable if intake-to-shoot rate is high.
- An adjustable hood lets one flywheel cover many distances; pair with turret+shooter code so aiming is automatic.
- Reliability is a process: they publish per-match pit checklists and run a role-specialized crew.
- Keep the climber cheap (lightweight L1) so mass/time goes to the scoring mechanisms that drive EPA.

**CAD release:** <https://www.chiefdelphi.com/t/team-341-miss-daisy-2026-cad-code-and-documentation-release/521088> · CD CAD/code/binder release · Reveal (Miss Daisy XXIV) · Robot code (public)

## Two-robot program: 'Asura' dual 720-deg twin-turret shooter, and 'Sandstorm' a 254/1678-style 4-wide dumper

|                                                                                                                                                                                                             |                                                                                                            |
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| <b>SHOOTER</b><br>Asura: TWO independent 720-degree turrets, each its own flywheel (~10 BPS combined). Sandstorm: 4-ball-wide top-level dumper/shooter (~19 BPS). Motor packaging iterated across test bots | <b>INTAKE</b><br>Full-width roller intake feeding the indexer; intake-deploy gearbox added last            |
| <b>INDEXER / HOPPER</b><br>Asura: ~45-ball hopper. Sandstorm: ~75-ball hopper. Concluded early they'd be 'indexer-limited' and hunted the fastest indexer concepts                                          | <b>DRIVETRAIN</b><br>CTRE swerve, Kraken X60 drive+steer, CANcoder, 7.67:1, 1.95in wheels, ~4.0 m/s        |
| <b>ENDGAME</b><br>Climber not detailed (two-robot program).                                                                                                                                                 | <b>AUTONOMOUS</b><br>Concluded indexing throughput was the binding constraint; auto routines not detailed. |
| <b>VISION</b><br>3x Limelight (left/center/right) via MegaTag                                                                                                                                               | <b>PUBLIC CAD / DESIGN</b><br>CAD + code for BOTH robots (Onshape/GitHub)                                  |

## LESSONS FOR STUDENTS

- Prototype the game piece FAST and find your binding constraint: they proved intake/shoot were easy and that INDEXING throughput was the real limit.
- Mine prior art - they studied 254 & 1678 plus RI3D/Open Alliance before committing.
- A dual-720-turret robot is buildable by a young team if you stage it (test bot first, then turret controls + tensioning).
- Same-team A/B study: directly compare their dual-turret vs dumper design choices.

**CAD release:** <https://www.chiefdelphi.com/t/team-9072-tigerbots-2026-cad-and-code-release/521421> · [CD CAD + code release \(both bots\)](#) · [Team robot page](#) · [Robot code \(public\)](#)

## High-quality fabrication: single shoot-on-the-move turret + spindexer on a DIY swerve (82% win rate)

|                                                                                                                                                                                                                                                                               |                                                                                                         |
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| <b>SHOOTER</b><br>Single turret with a 'breadbox' adjustable hood; 3D-printed turret bearing; inverted motor mount; built for shoot-on-the-move                                                                                                                               | <b>INTAKE</b><br>Roller intake (hardened States->Worlds; no major redesign)                             |
| <b>INDEXER / HOPPER</b><br>Spindexer feeding the turret; intentionally flexes out of the way for battery changes (ergonomics); no drum                                                                                                                                        | <b>DRIVETRAIN</b><br>DIY in-house 'Third Coast' swerve on a Chrome-Moly frame; many parts 3D printed    |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                                                                                                                                                                                                        | <b>AUTONOMOUS</b><br>Built for shoot-on-the-move (single 'breadbox'-hood turret); fast-iterated vision. |
| <b>VISION</b><br>FOUR cameras (two on each corner) on TWO Orange Pis (tucked under the batteries) running their custom 'Walleye' AprilTag software (in use since AprilTags launched); used heavily to correct the gyro/odometry in this odometry-driven game [BtB transcript] | <b>PUBLIC CAD / DESIGN</b><br>Onshape (2026 robot)                                                      |

## LESSONS FOR STUDENTS

- DIY swerve design - their full mechanical design doc (Third Coast) is public.
- 3D-printed turret bearing + inverted motor mount - clever, copyable packaging.
- A spindexer that flexes for serviceability - design for repair / battery access.
- Iterate for robustness (intake hardening) without scope creep.

**Onshape CAD:** <https://cad.onshape.com/documents/a4cd67d32bba9211459890b2> · [CD 2026 CAD](#) · [Onshape CAD](#) · [Third Coast swerve design doc](#) · [GitHub](#) · [Behind the Bumpers \(full transcript\)](#)

## Full-width turret shooter, bang-bang flywheel, shoot-on-move; 254/2910-style code

|                                                                                                                                                                                                                                                    |                                                                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Full-width turret; 6328-style bang-bang flywheel wrapped in a state machine; shot interpolation from data; SOTM capped ~0.5 m/s (rising); time-of-flight tracking; passing/shooting auto-selected by robot position + match time | <b>INTAKE</b><br>Independent intake state machine (full-width feed); mechanical detail not fully specified                                                                 |
| <b>INDEXER / HOPPER</b><br>Feeds the turret; managed within the superstructure state machine                                                                                                                                                       | <b>DRIVETRAIN</b><br>Swerve; PathPlanner autos (kept partly command-based to avoid state-machine conflicts)                                                                |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                                                                                                                                                                             | <b>AUTONOMOUS</b><br>Shoot-on-the-move (~0.5 m/s, rising) with time-of-flight tracking; passing vs shooting auto-selected by robot position; 254/2910-style state machine. |
| <b>VISION</b><br>Limelight + AprilTags (AprilTag field layout; a lazy-init bug once cost a loop overrun + ~5 hrs at field)                                                                                                                         | <b>PUBLIC CAD / DESIGN</b><br>Onshape CAD + public GitHub                                                                                                                  |

## LESSONS FOR STUDENTS

- Superstructure/state-machine control (a la 2910/254 ControlState) is more flexible & testable than scattered commands.
- Adopt proven patterns wholesale: 6328 bang-bang + AdvantageKit IO layer, 254 ControlState for isolating subsystems to test.
- A CI pipeline (build/format/sim gates) + an AI first-pass review scales code quality - caught ~35 bugs incl. a missing switch break that would have broken aiming.
- The real bottleneck is PR review / learning / robot test time, not code generation.

**Onshape CAD:** <https://cad.onshape.com/documents/91500087e03b9e406a527d6d> · [CD build thread](#) · [Onshape CAD](#) · [GitHub RainMaker26](#)

## College-mentored team that built and CAD-released TWO distinct competition robots

|                                                                                                                                                                        |                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>V2 TURNOVER is a turret shooter with a belt 'kicker' that feeds/compresses the ball into the flywheel (they stayed turret while many moved to drums) | <b>INTAKE</b><br>Kraken X60 FOC roller intake (integrated 'turnover' architecture)               |
| <b>INDEXER / HOPPER</b><br>Kraken X60 FOC rollers route fuel through the 'turnover' path into the turret                                                               | <b>DRIVETRAIN</b><br>CTRE swerve, Kraken X60 FOC                                                 |
| <b>ENDGAME</b><br>Climber not detailed (two robots: V1 'Doomspiral' + V2 'TURNOVER').                                                                                  | <b>AUTONOMOUS</b><br>Auto routines not detailed; the study value is the V1-vs-V2 A/B comparison. |
| <b>VISION</b><br>3x Limelight 4 (shooter-mounted, climber, intake positions), MegaTag2 via their GompeiLib                                                             | <b>PUBLIC CAD / DESIGN</b><br>Onshape - both robots (Doomspiral V1 + TURNOVER V2)                |

## LESSONS FOR STUDENTS

- Comparing two of one team's own robots (V1 vs V2) is a rare A/B design study - what changed and why.
- Turret-vs-drum trade study: they went turret while the field trended to drums - good design-decision reasoning.
- A belt-kicker that compresses the ball into the flywheel is a clean, copyable feeding concept.

**Onshape CAD:** <https://frc190.onshape.com/documents/ba8a365713d40fc24cf0b54f> · [CD 190 CAD release \(both robots\)](#) · [TURNOVER showcase](#) · [Onshape \(TURNOVER\)](#) · [Robot code \(2k26\)](#)

## Swappable-hopper quad-lane shooter ('Noah's Ark'): tall ~100-fuel config or short trench config, swapped in <5 min

|                                                                                                                                                                                          |                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Fixed (non-turret) QUAD launcher - four lanes, each on its own Kraken X60, spin-up under 0.25 s; kept under 22 in tall to pass under the trench.                       | <b>INTAKE</b><br>'Lintake' - a 5+ fuel-wide LINEAR intake with a powered kicker roller.                                                                                     |
| <b>INDEXER / HOPPER</b><br>Swappable hopper (~100-fuel tall config OR a short trench config, swapped in under 5 min via 8 bolts) feeding six powered floor rollers with zero dead zones. | <b>DRIVETRAIN</b><br>23.5 x 31.5 in base on SDS MK5n swerve (geared to R2), 7075 punch tube.                                                                                |
| <b>ENDGAME</b><br>Designed around scoring rather than climbing the Tower (climber not a focus).                                                                                          | <b>AUTONOMOUS</b><br>Full-field localization + shooter automation for scoring autos; PathPlanner-based routines (binder details the autonomous + shot-automation sections). |
| <b>VISION</b><br>4x Limelight 4 (LL4) full-field localization.                                                                                                                           | <b>PUBLIC CAD / DESIGN</b><br>Full CAD (Onshape) + code (GitHub) + web technical binder                                                                                     |

### LESSONS FOR STUDENTS

- A swappable hopper that reconfigures tall-vs-trench between matches exploits the multi-configuration inspection rules - capacity when you can, low profile to clear the trench and beat defense.
- An 'Alphabot' pre-season testbed surfaced problems early (inaccurate drum shot, illegal start config) before the real build.
- Four independent shooter lanes (one Kraken X60 each) give a high fire rate with sub-0.25 s spin-up.

**Onshape CAD:** <https://cad.onshape.com/documents/d7e50bc5f2f3accc3e502b3e/w/6c7c691839b17298be38e609/e/a406643d94e0f4b9284c165c> · [CAD/Code/Binder release \(CD\)](#) · [Technical Binder](#) · [CAD \(Onshape\)](#) · [Code \(GitHub\)](#) · [Statbotics](#)



## 'Infrared': turreted hood launcher, extending roller-bed intake, ball-tower indexer; 5-robot dev line (AM/XM/PM/FM/Photon); swerve

|                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Turret-mounted launcher with hood. Per the blog motor budget: turret on a Kraken X44, 2x Kraken X60 launcher flywheel, hood on an X44 or servos; Limelight rides on the turret (visible in the FM CAD next to a Talon FX-controlled motor).                                                                                                                                               | <b>INTAKE</b><br>12 in extending roller-bed intake (XM sketches): rack-gear extension (linear or slightly curved), single 2 in roller with a fold-down kicker bar, deploy motors stationary near the front frame rail. Final geometry sits 5 deg over the swerve module - the smallest angle that still compresses FUEL into the carpet (per mentor Q&A). |
| <b>INDEXER / HOPPER</b><br>Roller bed feeds two-ball inputs through 2-to-1 gates into a single ball tower that feeds the turret; lower powered rollers plus top rollers vector the balls. Hopper walls are laser-cut polypropylene (used as a laserable, Loctite-safe polycarb substitute). They also published a 4-lane 'sideways spindexer' concept (4x throughput idea) they considered for the rebuild. | <b>DRIVETRAIN</b><br>Swerve. AM (alpha) ran an existing MK4i base with MK5n considered for space claims; later machines refined the same footprint (final module not explicitly stated in the blog).                                                                                                                                                      |
| <b>ENDGAME</b><br>Climber not detailed in the released dev-line CAD.                                                                                                                                                                                                                                                                                                                                        | <b>AUTONOMOUS</b><br>Dev-line approach (AM/XM/PM/FM): final vision/auto decisions deliberately deferred until after seeing the game at real events.                                                                                                                                                                                                       |
| <b>VISION</b><br>Limelight mounted on the turret; blog discusses aiming via odometry vs turret-camera tx/ty, with AprilTags across the field providing pose updates.                                                                                                                                                                                                                                        | <b>PUBLIC CAD / DESIGN</b><br>CAD for all 5 robots (Onshape)                                                                                                                                                                                                                                                                                              |

### LESSONS FOR STUDENTS

- Their AM/XM/PM/FM/Photon 5-robot cadence intentionally delays final design decisions until AFTER seeing the game played at real events - architecture as a season-long iteration, not a kickoff bet.
- Publishing every machine's CAD (alpha through final) gives a rare public record of how one architecture matures across four rebuilds in a single season.
- Small geometric insights matter: the intake angle is exactly the minimum (5 deg) that clears the swerve module while still pressing FUEL into the carpet.
- Material pragmatism: polypropylene hopper walls because they laser-cut safely (polycarb doesn't) and take Loctite - process constraints driving material choice.

**CAD release:** <http://2026cad.spectrum3847.org/> · OA Build Blog (CD, 416 posts) · 2026 CAD - all 5 robots · Statbotics

## 'Miami': MK5n swerve, Madtown(1323)-style roller intake v2, short spindexer indexer, rear Limelight mount; launcher detail in CAD

|                                                                                                                                                                                                                        |                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Launcher specifics not documented anywhere public - the top-rear roller launcher assembly is visible in the Fusion model; read ratios/motors from the CAD.                                           | <b>INTAKE</b><br>'Madtown_Intake_v2' (92-part subassembly): a roller ground intake in the style popularized by 1323 Madtown Madness, iterated to v146 in their Fusion history. |
| <b>INDEXER / HOPPER</b><br>'Spindex_Short' - a compact spindexer, by far the largest subassembly in the model (397 parts). Combined with the intake naming, the architecture is intake -> short spindexer -> launcher. | <b>DRIVETRAIN</b><br>SDS MK5n swerve base ('Swerve_Base_mk5n_2026', 93 parts) on a sheet-metal chassis with riveted side panels.                                               |
| <b>ENDGAME</b><br>Climber not documented in the public CAD.                                                                                                                                                            | <b>AUTONOMOUS</b><br>Rear Limelight mount modeled for vision coverage; auto routines not public.                                                                               |
| <b>VISION</b><br>Limelight on a dedicated rear mount ('back_LimeLight_Mount'); two camera-FOV volumes ('FOV Off') are modeled in the assembly to plan vision coverage.                                                 | <b>PUBLIC CAD / DESIGN</b><br>Full robot CAD (Fusion 360 / A360)                                                                                                               |

### LESSONS FOR STUDENTS

- Modeling camera FOV volumes directly in the robot assembly is a cheap way to verify vision coverage before building anything.
- Naming a subsystem after its inspiration ('Madtown intake') and iterating it 140+ versions shows healthy borrow-then-refine engineering.
- A 'short' spindexer keeps the proven anti-jam architecture while reclaiming vertical space - worth comparing against full-height spindexers like 1756's previous seasons or 3847's concept studies.
- A complete, well-organized Fusion 360 release is rare (most teams publish Onshape) - valuable for Fusion-based teams wanting a reference structure.

**Team site:** <https://a360.co/4c8kxxw> · Full CAD (Fusion/A360) · Spectrum CAD Collection · Statbotics

## First-year Open Alliance team that won the Bluff Country Regional with turret + spindexer + adjustable hood

|                                                                                                                                                                       |                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Hooded shooter with an adjustable hood; the hood was a key iteration point - broke two before settling on a wide-bottomed, more rigid design        | <b>INTAKE</b><br>Roller intake (prototyped alongside shooter/spindexer/hopper using KrayonCAD conceiving)                      |
| <b>INDEXER / HOPPER</b><br>Spindexer feeding the shooter, plus a hopper; tuned in code for spindexer rotation/feeding                                                 | <b>DRIVETRAIN</b><br>Swerve, Kraken X60 drive + Kraken X44 steer, top-mount 'brainpan' electronics; built to shoot-on-the-move |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                                                                                                | <b>AUTONOMOUS</b><br>Auto routines not detailed (first-year Open Alliance Regional winner).                                    |
| <b>VISION</b><br>Limelight for turret aiming + pose; code adds voltage/supply/torque inputs and tunes intake, feed, shoot, spindexer, hood angle, and turret position | <b>PUBLIC CAD / DESIGN</b><br>Onshape (2026 robot)                                                                             |

### LESSONS FOR STUDENTS

- A hood failure-and-redesign cycle (broke two, then widened the base) - a concrete structural-iteration lesson.
- Design prototypes for adjustability/versatility from one set of core parts (sliding-hood compression plate).
- Limelight-based turret aiming with code-tuned mechanism setpoints - an accessible vision approach for a mid-tier team.
- A first-year Open Alliance team documenting its way to a regional win.

**Onshape CAD:** <https://cad.onshape.com/documents/b3b80a0c9175b5a0ee25425b> · CD build thread · Onshape CAD · Behind the Bumpers

### Tall spindexer-turret architecture with a full-width linear (rack-and-pinion) intake; Iowa Regional winner

|                                                                                        |                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Flywheel shooter on a 360+ degree turret atop a tall spindexer tower | <b>INTAKE</b><br>Full-width LINEAR intake driven by a 10-DP rack and pinion (tooth profile modified to cut with a 1/8in end mill, inspired by 3467/6328); racks went polycarb -> 2x .2in MAXComposite + 1x .1in aluminum for tooth strength |
| <b>INDEXER / HOPPER</b><br>Tall spindexer feeding the turret; ~80-fuel hopper          | <b>DRIVETRAIN</b><br>Swerve (module unconfirmed)                                                                                                                                                                                            |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                 | <b>AUTONOMOUS</b><br>Auto routines not detailed (Iowa Regional winner).                                                                                                                                                                     |
| <b>VISION</b><br>Unconfirmed (turret-aim code released)                                | <b>PUBLIC CAD / DESIGN</b><br>Full CAD release (Onshape) - first comp robot fully in Onshape                                                                                                                                                |

#### LESSONS FOR STUDENTS

- A rack-and-pinion linear intake gives a controlled straight-line deploy; 10-DP teeth modified for a 1/8in end mill make them CNC-able.
- Material selection is iterative: polycarb rack teeth sheared, so they moved to a MAXComposite + aluminum laminate - test under real load.
- Switching to Onshape pays off with parametric top-down design + FeatureScript despite transition hiccups.
- Plan for wear: turret wire sleeving wore quickly and became a per-event replacement.

**CAD release:** <https://www.chiefdelphi.com/t/team-3928-team-neutrino-2026-cad-release/521183> · [CD CAD release](#) · Statbotics

### Adjustable-hood dual-motor launcher + hopper feeder; high-volume scorer; qualified Worlds

|                                                                                                                                                              |                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Adjustable-hood launcher, two flywheel motors, linear-actuator hood read by an absolute encoder; high throughput (~260 fuel/match claimed) | <b>INTAKE</b><br>Floor intake feeding the hopper (works with hopper to empty fuel quickly)                                                     |
| <b>INDEXER / HOPPER</b><br>Hopper + feeder delivering fuel to the dual-motor launcher                                                                        | <b>DRIVETRAIN</b><br>Swerve; drivetrain assembled early; vision still in progress late-Feb                                                     |
| <b>ENDGAME</b><br>Side L1 climb - the CG offset from the hook needs ~0.6 friction or a dedicated gripper.                                                    | <b>AUTONOMOUS</b><br>Adjustable-hood dual-motor launcher with an absolute-encoder hood for repeatable shot angles; auto routines not detailed. |
| <b>VISION</b><br>Unconfirmed (integration in progress during build)                                                                                          | <b>PUBLIC CAD / DESIGN</b><br>Onshape CAD + GitHub (Open Alliance blog)                                                                        |

#### LESSONS FOR STUDENTS

- A linear-actuator hood with an absolute encoder gives repeatable, tunable shot angles for variable distance.
- Side L1 climbing creates a tricky moment - CG offset from the hook needs ~0.6 friction or a dedicated gripper.
- Dual launcher motors must be monitored - a single over-current motor silently halves your shooting (watch current draw).
- Document openly (GitHub/Onshape/blog) to keep iteration disciplined through a Worlds-qualifying season.

**CD thread:** <https://www.chiefdelphi.com/t/2582-pantherbots-build-blog-open-alliance-2026/509632> · [CD build blog](#) · [Reveal 'Cobra'](#)

## 'HAILSTORM': compact turret shooter fed by a spindexer; motor + single absolute-encoder turret; Kotlin code; swerve

|                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Turret-mounted launcher (described as a compact turret assembly). Per the thread, the team weighed a 'dye rotor' but rejected it as too complex and chose the turret/spindexer path for simplicity with a high ceiling. Flywheel motor/ratio specifics not stated in the thread - read from the Onshape model. | <b>INTAKE</b><br>Ground intake; the team noted a mid-season intake redesign that delayed their practice-field time. Specifics not detailed in the thread - see CAD. |
| <b>INDEXER / HOPPER</b><br>Spindexer feeding the turret. The team chose a spindexer over a dye rotor as the simplest solution with a high ceiling. (The cone was finished off-CAD with cat-tongue grip-tape strips, heat-shrunk to conform for extra grip.)                                                                      | <b>DRIVETRAIN</b><br>Swerve (module/gearing not stated in the thread - see CAD).                                                                                    |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                                                                                                                                                                                                                                                           | <b>AUTONOMOUS</b><br>Turret zeroing via a motor (relative) encoder + one absolute encoder; auto routines not detailed.                                              |
| <b>VISION</b><br>Not named in the thread - the turret aims via encoders rather than a stated camera; treat as Unconfirmed and check the CAD/code repo.                                                                                                                                                                           | <b>PUBLIC CAD / DESIGN</b><br>Full robot CAD (Onshape) + code (GitHub, Kotlin)                                                                                      |

### LESSONS FOR STUDENTS

- Pragmatic turret zeroing: the motor (relative) encoder handles continuous tracking, and a single absolute encoder geared ~5:1 supplies a re-zero value while shooting; closest-region logic plus manual re-zero buttons cover encoder drift - a cheap alternative to a full CRT/absolute setup they couldn't tune in time.
- Archetype selection as risk management: they rejected a dye rotor for a spindexer specifically because it was the simplest path with a high performance ceiling.
- A public Kotlin codebase is a useful non-Java reference for teams exploring JVM languages.
- Recovery story worth reading: early power-consumption issues, then a hardening/tuning focus that made them one of their team's best-ever robots by DCMF.

**Onshape CAD:** <https://cad.onshape.com/documents/c527fb2ecdd377cbb954f474/w/b89f138221c8d9223c28e242/e/c4daa58f8630b2239d8ee484> · [CAD & Code release \(CD\)](#) · [CAD \(Onshape\)](#) · [Code \(GitHub\)](#) · [Statbotics](#)

## Turret spindexer shooter, fully in-house manufactured, fast-iterated vision/SOTM; Midwest Regional winner

|                                                                                                              |                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Turret-mounted flywheel with shoot-on-the-move                                             | <b>INTAKE</b><br>Roller intake with a quick-change roller (a bent roller is a fast pit swap); stiffened + de-jammed after the first regional |
| <b>INDEXER / HOPPER</b><br>Spindexer (cone geometry inspired by 4145)                                        | <b>DRIVETRAIN</b><br>Swerve (module unconfirmed)                                                                                             |
| <b>ENDGAME</b><br>Climber present but iterated for reliability (they considered removing it between events). | <b>AUTONOMOUS</b><br>Two Limelight 4s + pose + shoot-on-the-move brought online in ~8 days between events.                                   |
| <b>VISION</b><br>2x Limelight 4 for AprilTag pose + shoot-on-the-move; stood up in ~8 days mid-season        | <b>PUBLIC CAD / DESIGN</b><br>Full CAD release (Onshape) - first Onshape season, 100% in-house parts                                         |

### LESSONS FOR STUDENTS

- Vision is a software sprint you can win late: two Limelight 4s + pose + SOTM working in ~8 days between events.
- Design for fast field repair - a quick-change intake roller turns a bend into a pit swap, not a dead robot.
- Iterate event-to-event with a clear list (stiffen intake, fix jamming, even remove the climber for reliability).
- Borrowing a subsystem detail (spindexer cone from 4145) beats reinventing it.

**CAD release:** <https://www.chiefdelphi.com/t/frc-48-team-e-l-i-t-e-xtremachen-29-2026-cad-release/521258> · [CD CAD release](#) · [Statbotics](#)

## High-capacity hopper shooter known for big event-to-event upgrades; ~80-ball usable storage

|                                                                                                           |                                                                                                         |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Flywheel shooter (turret detail unconfirmed); continuous upgrades each event            | <b>INTAKE</b><br>Roller intake (motors unconfirmed); throughput slows once the hopper is heavily loaded |
| <b>INDEXER / HOPPER</b><br>Large hopper: ~80 balls before fuel weight bogs the intake, ~90-95 at overflow | <b>DRIVETRAIN</b><br>YAGSL swerve, Kraken X60 drive + Kraken X44 steer, CANcoder, navX                  |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                                    | <b>AUTONOMOUS</b><br>Auto routines not detailed; large event-to-event upgrades.                         |
| <b>VISION</b><br>3x Limelight (left/right/front), MegaTag pose estimation                                 | <b>PUBLIC CAD / DESIGN</b><br>CAD + code release (Onshape)                                              |

### LESSONS FOR STUDENTS

- Add-on Delrin 'skis' under the drivebase help the robot ride over the field bump and shield swerve motors - a cheap fix for a known hazard.
- Know your REAL capacity limit (~80 usable) vs physical (90-95) - design to the usable number.
- Aggressive event-to-event upgrades drove a steep improvement curve.

**CAD release:** <https://www.chiefdelphi.com/t/1676-cad-release-2026/520302> · [CD CAD + code release](#) · [Team mechanical page](#) · [Released code \(practice chassis\)](#)

## Drum shooter (no turret) with a compression-tuned over-bumper intake + roller indexer

|                                                                                                                                                            |                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Drum shooter (team very satisfied with it); planned to grind the hood for a higher-arching, more accurate shot; SOTM not yet implemented | <b>INTAKE</b><br>Over-bumper intake whose compression was too high - couldn't copy another team's geometry, had to run their own compression tests; broke several times and was reinforced with aluminum plates |
| <b>INDEXER / HOPPER</b><br>Indexer rollers feeding the drum; planned smaller indexing rollers to raise throughput                                          | <b>DRIVETRAIN</b><br>Kraken X60 swerve (TalonFXS modules)                                                                                                                                                       |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                                                                                     | <b>AUTONOMOUS</b><br>Drum shooter; shoot-on-the-move not yet implemented (planned).                                                                                                                             |
| <b>VISION</b><br>4x PhotonVision cameras (back-left, back-right, side-left, side-right) - PhotonVision, not Limelight                                      | <b>PUBLIC CAD / DESIGN</b><br>Onshape CAD + GitHub (Open Alliance)                                                                                                                                              |

### LESSONS FOR STUDENTS

- Intake compression must be tuned to YOUR robot - copying another team's spacing doesn't transfer; run your own ball-compression tests.
- Design for durability: an intake that breaks repeatedly needs structural reinforcement, not just tuning.
- Smaller indexing rollers can raise throughput - small geometry changes meaningfully affect ball flow.
- Borrow hopper-capacity ideas from top teams (passive net like 5687, vertical active expansion like 1678).

**CD thread:** <https://www.chiefdelphi.com/t/frc-2079-alarm-robotics-2026-rebuilt-build-thread/509762> · [CD build thread](#) · [GitHub](#) · [Robot code \(DeadShot\)](#)

### Small (17-student) team that built an ambitious turret shooter with a star-wheel intake

|                                                                                                                                                                                          |                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Turret: dual Kraken X60 on a 1:1 shooter axle, two SDS brass flywheels + a single 4in Fairlane 60A wheel; adjustable hood with a 1in hood roller geared 2:1            | <b>INTAKE</b><br>Over-bumper 4in compression wheels + star wheels (no kicker bar needed); deployed by dual mirrored rack-and-pinion (racks = 4 epoxied laminations of .120 polycarb, PETG pinions) |
| <b>INDEXER / HOPPER</b><br>Roller hopper with a passive funnel + a kicker that flicks balls into the shooter (254-2017 style); chose roller hopper over spindexer after prototyping both | <b>DRIVETRAIN</b><br>Swerve, Kraken X60 drive, 4.2:1, 38mm wheels, ~5.5 m/s, 0.55m track width                                                                                                     |
| <b>ENDGAME</b><br>Climber not detailed (small 17-student team).                                                                                                                          | <b>AUTONOMOUS</b><br>Auto routines not detailed.                                                                                                                                                   |
| <b>VISION</b><br>4x Limelight (one..four), MegaTag2                                                                                                                                      | <b>PUBLIC CAD / DESIGN</b><br>Onshape (2026 document)                                                                                                                                              |

#### LESSONS FOR STUDENTS

- A star-wheel + compression-wheel over-bumper intake can work WITHOUT a kicker bar.
- Rack-and-pinion intake deploy with laminated-polycarb racks - fabrication on a budget.
- Roller-hopper vs spindexer: they prototyped both and documented why they picked rollers.
- Honest self-critique (3 lb of brass flywheel) models good engineering process for students.

**Onshape CAD:** <https://cad.onshape.com/documents/589237fcf2364719e491ed4b> · [CD Open Alliance thread](#) · [Onshape CAD](#) · [Website](#) · [Robot code](#)

### Wide-intake high-capacity turret shooter, fully student-built; reached Newton Division

|                                                              |                                                                                                       |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Flywheel turret shooter, ~19 BPS sustained | <b>INTAKE</b><br>5-balls-wide intake                                                                  |
| <b>INDEXER / HOPPER</b><br>Large hopper, ~75-ball capacity   | <b>DRIVETRAIN</b><br>Swerve at an 'R2' (MAXSwerve-class) ratio; max frame perimeter, max-weight robot |
| <b>ENDGAME</b><br>Climber not detailed in the release.       | <b>AUTONOMOUS</b><br>Auto routines not detailed; ~19 BPS sustained throughput.                        |
| <b>VISION</b><br>Unconfirmed                                 | <b>PUBLIC CAD / DESIGN</b><br>CAD + Technical Binder (Onshape)                                        |

#### LESSONS FOR STUDENTS

- A wide intake (5 across) + a big ~75-ball hopper makes a high-throughput cycle machine - width and capacity are force multipliers.
- Fully in-house, student-led manufacturing is achievable; pair it with a written technical binder so the reasoning survives to next year.
- Pick a known-good swerve ratio ('R2') instead of over-customizing, and spend the saved effort on scoring mechanisms.

**CAD release:** <https://www.chiefdelphi.com/t/steampunk-1577-2026-cad-and-technical-binder-release/521240> · [CD CAD + tech binder release](#) · [Statbotics](#)

## 720-deg turret shooter with a 3D-printed spindexer and a 1678/27-style slapdown intake; mid-season rebuild to add a hopper

|                                                                                                                                                            |                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Flywheel on a 720-degree turret rotating on a 7in X-Contact bearing; praised for tight turret wire management + ball-exclusion shielding | <b>INTAKE</b><br>'27/1678-inspired' slapdown intake; later swapped to a rebuilt 'Buckeye' 4-bar intake mid-season |
| <b>INDEXER / HOPPER</b><br>3D-printed spindexer; rebuilt between Wk3 and Wk5 to move the intake to the short side, freeing room to add a hopper            | <b>DRIVETRAIN</b><br>Swerve, 32x23.5in base                                                                       |
| <b>ENDGAME</b><br>Climber not detailed; a mid-season rebuild added a hopper.                                                                               | <b>AUTONOMOUS</b><br>Auto routines not detailed.                                                                  |
| <b>VISION</b><br>Unconfirmed                                                                                                                               | <b>PUBLIC CAD / DESIGN</b><br>CAD release + reveal video (Onshape)                                                |

### LESSONS FOR STUDENTS

- Don't fear a mid-season rebuild: moving the intake from the long side to the short side opened room for a hopper and 'significantly improved' offense.
- Borrow proven geometry - their slapdown intake is openly modeled on 1678/27, a fast path to a working mechanism.
- A 720-degree turret on a big X-contact bearing needs disciplined wiring + a ball-exclusion shield.
- 3D-printed spindexers are a low-cost, iterable storage option when you can't machine custom parts.

**CAD release:** <https://www.chiefdelphi.com/t/team-5811-the-bonds-2026-robot-cad-release/520901> · [CD CAD release](#) · Statbotics

## 720-deg passive central front turret; alternating-direction polybelt serializer; large hopper; MK4n swerve sized to fit under the trench

|                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Shooter rides a central front turret with 720 deg of travel achieved via a passive mechanism (no re-zeroing); wheel/motor specifics not stated in the thread - see CAD.                                                                                                                                                                                                                               | <b>INTAKE</b><br>Ground intake feeding the large hopper (specifics not detailed in the build thread - see CAD).                                  |
| <b>INDEXER / HOPPER</b><br>Center belted serializer: paired polybelt wheel stacks spun in ALTERNATING directions so FUEL cannot interlock and jam, funneling the wide hopper into one stream. Reached via 4+ prototypes - a spindexer (too sensitive to gravity/geometry) and a two-wheel track (balls interlocked) were rejected; V1 (top belt only) ran the Oklahoma and Magnolia regionals before the final version. | <b>DRIVETRAIN</b><br>SDS MK4n swerve; overall height constrained to drive under the trench, which motivated the low central turret architecture. |
| <b>ENDGAME</b><br>Climber not detailed in the public release.                                                                                                                                                                                                                                                                                                                                                           | <b>AUTONOMOUS</b><br>Auto routines not detailed.                                                                                                 |
| <b>VISION</b><br>Unconfirmed (not stated in the build thread).                                                                                                                                                                                                                                                                                                                                                          | <b>PUBLIC CAD / DESIGN</b><br>Full CAD (Onshape) + subsystem build thread                                                                        |

### LESSONS FOR STUDENTS

- Counter-rotating adjacent serializer belts is a cheap, mechanical anti-jam trick - the geometry physically can't wedge two balls.
- Their thread shows the full failure path (spindexer -> track -> belted serializer): documenting WHY prototypes failed is as instructive as the final design.
- A passive 720-deg turret gets shoot-anywhere coverage without slip rings or complex wiring - clever low-budget engineering.
- Packaging constraints (fit under the trench with MK4n modules) drove the whole architecture - constraints first, mechanism second.

**Onshape CAD:** <https://cad.onshape.com/documents/0f18097cc68763333bbf9187/w/ec4ab9bf696dddb823f66e6c/e/e6c97ba1f43ead9fa84d0197> · [CD build thread \(turret + serializer\)](#) · [Full CAD \(Onshape\)](#) · Statbotics

## Simplicity-first single shooter (NO turret), modular per-subsystem Onshape docs

|                                                                                                                                                                                              |                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Single fixed shooter (no second shooter, no turret) - chose one for accuracy, judged they can move balls fast enough; turret deferred to the offseason due to inexperience | <b>INTAKE</b><br>Over-bumper / floor intake (its own Onshape doc; outtake variants prototyped on YouTube) |
| <b>INDEXER / HOPPER</b><br>Dedicated indexer subsystem in its own Onshape doc feeding the single shooter                                                                                     | <b>DRIVETRAIN</b><br>Chassis subsystem in its own Onshape doc (swerve expected)                           |
| <b>ENDGAME</b><br>Climber not detailed (simplicity-first, no turret).                                                                                                                        | <b>AUTONOMOUS</b><br>Auto routines not detailed; one accurate fixed shooter by choice.                    |
| <b>VISION</b><br>Unconfirmed (no clear vision pipeline documented)                                                                                                                           | <b>PUBLIC CAD / DESIGN</b><br>Onshape - one doc per subsystem (Shooter/Indexer/Intake/Chassis/Climber)    |

## LESSONS FOR STUDENTS

- Scope to your team's experience: deliberately skipping a turret keeps a developing team reliable rather than over-reaching.
- One accurate shooter can beat two if you can feed balls fast enough - fewer mechanisms, fewer failure points.
- Splitting CAD into one Onshape doc per subsystem lets students work in parallel without conflicts.
- Defer ambitious mechanisms (turret) to the offseason as a learning project rather than risking the season.

Onshape CAD: <https://cad.onshape.com/documents/e139ac1c610559bc8767c8da> · [CD build thread](#) · [Onshape subsystem docs](#)

## 'Doubloon': WCP Competitive Concept-based scorer reworked for serviceability; PhotonVision

|                                                                                                                          |                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Based on the WCP Competitive Concept shooter (specifics not detailed in the release).                  | <b>INTAKE</b><br>Quick-change intake: every roller runs on standard hex shafts so a broken roller swaps in under a minute; geometry revised for more consistent pickup. |
| <b>INDEXER / HOPPER</b><br>Lightweight net hopper replacing the CC's heavy extendable hopper — lighter and more durable. | <b>DRIVETRAIN</b><br>Swerve, built from the WCP Competitive Concept base (module specifics unconfirmed).                                                                |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                                                   | <b>AUTONOMOUS</b><br>PhotonVision; auto routines not detailed.                                                                                                          |
| <b>VISION</b><br>PhotonVision — integrated for on-field targeting (not modeled in the CAD).                              | <b>PUBLIC CAD / DESIGN</b><br>Full CAD (Fusion 360)                                                                                                                     |

## LESSONS FOR STUDENTS

- Adapting a COTS concept (WCP Competitive Concept) and optimizing it for serviceability — quick-change hex-shaft rollers — is a legitimate, time-efficient design path.
- Replacing a complex extendable hopper with a simple net cut weight and improved durability: simplicity as a design win.

Team site: <https://a360.co/3NHiaZ3> · [CD CAD release](#) · [Statbotics](#)



## Modest on-field results but outstanding, deeply-documented vision + turret engineering

|                                                                                                                                                                                                                             |                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>SHOOTER</b><br>Two-motor single-wheel flywheel using 4in ThriftyBot urethane wheels at ~1in compression; adjustable hood 20.0-57.5 deg                                                                                   | <b>INTAKE</b><br>Roller intake; added a second Kraken X60 which greatly sped up intaking and reduced jams                  |
| <b>INDEXER / HOPPER</b><br>Polycarb-roller hopper slanted toward the uptake (grip via hockey tape / textured TPU); 254-style turret wiring; turret sensed by an effective 1:1 encoder over 360 deg                          | <b>DRIVETRAIN</b><br>Thrifty Narrow Swerve via the CTRE Swerve Project Generator; uses YAMS (Yet Another Mechanism System) |
| <b>ENDGAME</b><br>Climber not detailed in the release.                                                                                                                                                                      | <b>AUTONOMOUS</b><br>Multi-camera PhotonVision on dual Orange Pi 5 (deeply documented vision); auto routines not detailed. |
| <b>VISION</b><br>PhotonVision 2026 on two Orange Pi 5 coprocessors (load split across ~4 ThriftyCams); 3 cams on the shooter platform for localization + 1 on the turret for hub alignment; custom AprilTag mounting plates | <b>PUBLIC CAD / DESIGN</b><br>Open Alliance build thread (CAD/links in-thread)                                             |

### LESSONS FOR STUDENTS

- Multi-camera PhotonVision on dual Orange Pi 5 - splitting cameras across coprocessors, MJPEG-vs-YUYV trade-offs, per-Pi hostnames (a thorough vision study).
- Separate localization cameras (chassis) from a targeting camera (turret) - a clean architecture pattern.
- 254-style turret wiring + a 1:1 absolute encoder for a 360-deg turret.
- Adding a 2nd intake motor to cure jams - a simple, high-impact reliability fix. (Study for vision/turret, not on-field results.)

**CD thread:** <https://www.chiefdelphi.com/t/frc-157-the-aztechs-2026-build-thread-open-alliance/509425> · [CD build thread](#) · [Statbotics](#)

**Sources:** Chief Delphi (reveals & Open Alliance build threads), David's Design Server / [frcdesign.org](http://frcdesign.org) (robot-pics-2026 & mechanism-examples), team CAD/code releases (Onshape, GrabCAD, GitHub), FUN "Behind the Bumpers", and Statbotics EPA & records (2026). "Unconfirmed" means a detail was not publicly published, not that it is absent.